

BALANCED HARVESTING EFFECTS: SIZE SPECTRUM MODELLING OF PRODUCTION-PROPORTIONAL FISHING IN THE CELTIC SEA

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DECLARATION OF AUTHORSHIP

Except where acknowledged, the work reported in this Report is my own. My supervisor, Jon Pitchford, provided the conventional support Gustav Delius provided training and advice on size spectrum modelling, its interpretation, and provided specific implementation of biological diffusion processes and age-length matching methodology to the mathematical modelling framework. James Rimmer provided training on *mizer* modelling, especially concerning the tuning of single species before incorporation into a multispecies model. Valentina Lauria, Angela Martin, and Paula Silvar provided the species diets as well as biomass estimates of the Celtic Sea. Richard Law provided extensive discussion concerning the balanced harvest literature and areas of further research. I am grateful for the boarfish age and length data provided by Maria Inês Silva and Ana Rita Vieira via their extensive collection of data and processing techniques for accurate age identification.

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1. ABSTRACT

Historical reliance on single-species Maximum Sustainable Yield (MSY) frameworks with strong size selectivity has concentrated fishing mortality on large predators, destabilizing marine ecosystem structure. Species- and size- level balanced harvesting (ssBH), in which fishing mortality scales proportionally with dynamic biomass production, has been proposed as an ecosystem-based alternative capable of sustaining high yields while preserving natural size spectra. However, concerns remain that balanced harvesting could also accelerate ecosystem degradation.

This study uses a dynamic multispecies size-spectrum model, *mizer*, to evaluate the ecological and food-provisioning consequences of transitioning from the current fishing regimes to hybrid and fully implemented ssBH regimes in the Celtic Sea across a range of fishing intensities and management weightings. The analysis examines how ssBH influences ecosystem structure, yield distribution across the food web, and the trade-offs between conservation and fisheries production objectives.

Model results show that ssBH redistributes exploitation away from depleted demersal stocks toward highly productive small pelagic species, increasing aggregate yield while promoting the persistence of large fish within the ecosystem. Fully implemented ssBH regimes at low fishing intensities maximize total biomass and spawning stock biomass but reduce large-fish catch biomass, potentially limiting economic viability. In contrast, hybrid ssBH configurations with intermediate management weightings better reconcile competing objectives by maintaining adult biomass while improving economic performance indicators.

Overall, this study provides a conceptual benchmark suggesting that production-proportional hybrid harvesting regimes can support ecosystem sustainability in the Celtic Sea as well as fishery objectives.

2. INTRODUCTION

The harvest of marine organisms for direct and indirect human consumption supports global food security, livelihoods, and economies (Costello et al., 2020; Selig et al., 2019). Ensuring these resources are sustainably managed remains a major challenge for fisheries management, which must balance efficient and maximum sustainable utilization of marine production (UNCLOS, 1982) with the preservation of stock structure and ecosystem functioning (CBD, 1992).

In the Celtic Sea, management has historically relied on Total Allowable Catches (TACs) derived from single-species Maximum Sustainable Yield (MSY) models. This framework is operationally coupled with strict size-selective limits designed to optimize yield-per-recruit by excluding juveniles from harvest (ICES 2025a, MMO, 2025; Froese et al., 2008, Froese et al., 2016a). This strategy is also heavily biased toward the removal of high-trophic-level species, utilizing gear restrictions to focus mortality on large-bodied predators with the highest market demand (Halliday and Pinhorn, 2002; Hall and Mainprize, 2005; McClenachan et al., 2012; Roberson et al., 2020). The prevailing logic suggests that focusing pressure on high-trophic adults maximizes revenue per fish caught while allowing lower-trophic forage fish to remain abundant as a food source (FAO, 2003; Froese et al., 2008; Froese et al., 2016 a; Pikitch et al., 2012).

Despite these intentions, the coupling of single-species MSY with heavy large size-selectivity distributes fishing in ways that deviate from the natural proportionality of the ecosystem, shifting communities towards states that are less stable, diverse, and capable of supporting productive fisheries in the long term (Pikitch et al., 2004; Worm et al., 2009; Garcia et al., 2012). Because large, older fish are highly fecund and buffer recruitment variability, their targeted removal decreases population resilience and increases instability (Birkeland and Dayton, 2005). This population-level instability can propagate through the food web, disrupting species interactions and triggering trophic cascades. Additionally, sustained pressure on high trophic level fish species shifts trophic structure toward smaller species, which may cause a reduction in ecosystem services, productivity, and loss of stabilizing top-down controls, potentially resulting in ecosystem collapse (Daan et al., 2005).

To address these imbalances, recent theoretical developments in fisheries biology have sought methods that ensure high sustainable yields while better maintaining ecosystem structure and function. One proposed alternative is balanced harvesting (BH), which aims to redistribute fishing mortality across the widest possible range of species and sizes in proportion to their natural productivity (Garcia et al., 2012). By aligning harvest with the inherent production of the entire size spectrum, BH is theorised to maintain ecosystem composition while still providing sufficient and

sustainable yields (Zhou et al., 2010; Kolding et al., 2016c; Garcia et al., 2012; Borges et al., 2016).

While many ecosystem models support the BH hypothesis (Law et al., 2012, 2016; Jacobsen et al., 2014; Zhou and Smith, 2017; Plank, 2018; Nilsen et al., 2020), the concept remains divisive. Critics argue that early models relied on simplifying assumptions-such as fixed plankton communities, a lack of predator-prey dynamics, and unnatural mortality/biomass relationships- which may have biased outcomes (Froese et al., 2016b). Furthermore, empirical data from unmanaged fisheries, such as those in small-African lakes, which have been speculated to follow BH regimes (Kolding et al., 2016b; Kolding and van Zwieten, 2011, 2014), have sparked further controversy. From the declining catch-per-unit-effort and low value species in the yields of these fisheries, critics fear that BH may be simply “fishing down the food web” under the guise of sustainable harvest (Tweddle et al., 2015; Froese et al., 2016b; Smith et al., 2011).

These critiques highlight the need for more realistic modelling frameworks to accurately illustrate the trade-offs of the regime, as well as emphasising the need for specific BH harvest rules, as simply spreading fishing across more species and sizes does not guarantee sustainability. To ensure BH remains within safe biological limits, it has been formalized through precise mathematical definitions of fishing mortality.

The two primary rules which have emerged are (a) applying fishing mortality proportional to productivity (Kolding et al., 2015) and (b) applying fishing mortality proportional to production (Andersen et al., 2016; Law and Plank, 2022). Following Zhou et al. (2019), production (P) is the total biomass produced by an ecological group per unit time, irrespective of its fate, while productivity (p) is the rate of production per unit of standing biomass (B), P/B . In contrast to the productivity rule, adopting the production rule directly addresses concerns regarding “fishing down the food web”; by introducing a density-dependent safeguard, fishing pressure automatically scales down as biomass declines, preventing the collapse of vulnerable stocks (Law and Plank, 2022; Plank, 2018; Rehren and Gascuel et al., 2020; Zhou et al., 2021). Furthermore, while species-level BH (sBH) balances mortality across taxa, species-and-size-level BH (ssBH) accounts for the individual body size. This distinction is critical, as ssBH more accurately reflects the size-structured nature of marine ecosystems and ontogenetic shifts in biomass flow, making it the most theoretically consistent application of balanced harvest principles (Zhou et al., 2019). Consequently, for the remainder of this study, BH refers specifically to fishing mortality proportional to production at the species-and-size level.

Despite these refinements, three significant knowledge gaps persist. First, the optimal proportion of production to extract from the system (known as the fishing intensity) is underexplored, as most studies utilize arbitrary intensities to match historical yield rather than identifying the system-wide optimal value (Law and Plank, 2022). Secondly, the ecosystem effects of “hybrid” strategies, where ssBH is implemented alongside existing status quo fishing, have not been intensively explored. Finally, the economic effects and practicalities of implementation, due to the potential of less high value species within catches in ssBH regimes and the limits in gear selectivity and management authorities, have yet to be formally addressed (Peters, 1986; Zhou et al., 2019).

While this last knowledge gap is largely beyond the scope of the present study, optimal fishing intensity and the effects of “hybrid” strategies will both be explored, using a dynamic multispecies size-spectrum model (mizer) to evaluate the conservation and food provisioning outcomes of ssBH compared to status quo fishing practices within the Celtic Sea. This region provides an ideal case study due to availability of data on species diet, biomasses, landings, and survey data (Lauria, 2016; ICES, 2025; European Commission, 2024, Silva et al., 2024). By exploring a parameter space of varying ssBH fishing intensities and management weightings, the transition from current practices to hybrid and whole-ssBH regimes will be examined.

Thus, the primary aim of this research is to determine if and to what degree gains in ecosystem health and food provisioning can be achieved through ssBH. These effects were evaluated by calculating total stock and spawning stock biomass, abundance, and the proportion and biomass of large fish in the system and the catch. Specifically, this study seeks to answer:

1. What are the ecological and provisioning effects of different fishing intensities within an entirely ssBH regime?
2. How does a hybrid or entirely ssBH regime perform compared to the status quo regime?
3. How does a transition to a hybrid or entirely ssBH regime redistribute yields across the food web?
4. What are the underlying ecological processes which drive these shifts when implementing hybrid or entirely ssBH regimes from the status quo?

3. METHODS

3.1. OVERVIEW

All dynamic multi-species size-spectrum models were developed and projected using the R package *mizer* (Scott et al., 2014). This framework was utilized due to its status as an established tool for evaluating the ecological effects and trade-offs of diverse fishing regimes (Blanchard et al., 2014; Pennino et al., 2025). Its robust integration of ecological and bioenergetic theory facilitated the realistic mechanistic modelling of individual growth, interspecific interactions, and biomass flux (Andersen et al., 2019).

In the present study, the standard modelling approach outlined in Scott et al. (2014) was substantially modified to enhance biological realism and streamline model construction. Key adjustments included a revised growth parameterisation (Section 3.4.1), preliminary tuning of species specific allometric background rates within a non-interacting model framework (Section 3.4 and 3.5), and an expanded empirical data-fitting process to ensure closer alignment with observed ecological patterns (Section 3.6).

Initially, a non-interacting, steady-state allometric status quo model, representing the average state of the Celtic Sea between 2012 and 2024, was constructed. Parameters were set to ecological estimates and tuned via data fitting methods (Table 2; Section 3.6). This data included a reference Ecopath model (Lauria, 2012; 2016), fisheries landings (Lauria, 2016; European Commission, 2024), survey length distributions (ICES, 2025b), maturity data (ICES, 2025b), age-length data (ICES, 2025b; Silva et al., 2024), a diet-matrix (Lauria, 2016), and stomach contents data (Delius, 2026).

Once calibrated, the status quo model had interspecific interactions enabled and was simulated forward in time to a steady-state. From this status-quo model hundreds of simulations exploring hybrid and entirely ssBH fishing regimes were projected to steady-state at which point the final metrics of each fishing scenario were evaluated. The structural stability and predictive reliability of conclusions were evaluated using a Morris and Regional sensitivity analysis (Morris, 1991; Pianosi et al., 2016). This approach allowed for a systematic exploration of the parameter space, ensuring that the study's conclusions regarding ssBH were robust to inherent uncertainties in the input data (Benoit et al., 2021; Douania et al., 2023; Cariboni et al., 2007).

The model construction process is visualized in Figure 1 with details of each step referencing the corresponding written section.

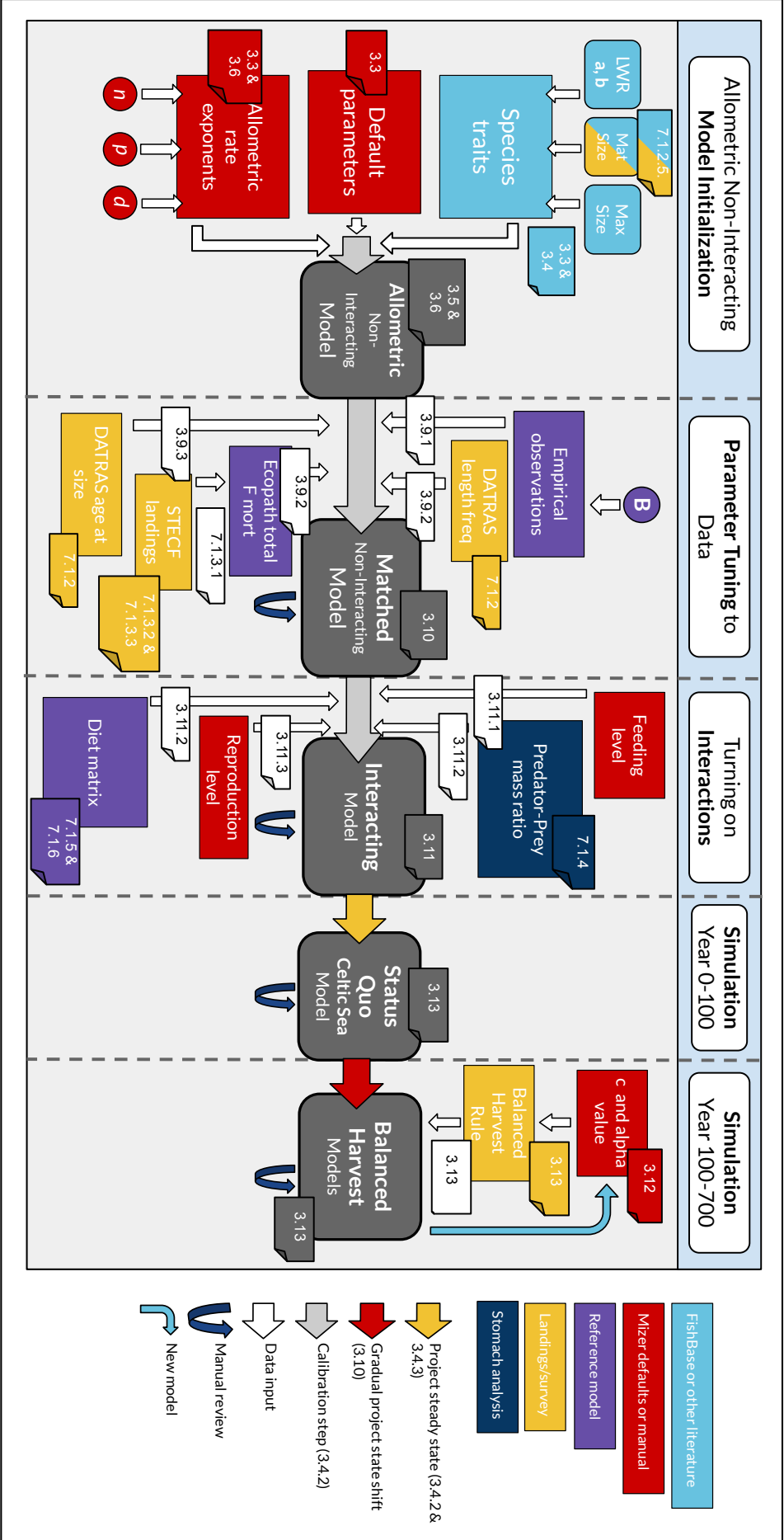


Figure 1: Schematic representation of the workflow and data sources used to build the mizer models with reference to the corresponding sections in the report noted as numerical values in parentheses or next to and colour matching the corresponding step. These sections include data processing techniques when corresponding with an input, integration of the data into the model parameterisation when corresponding with an arrow and represent the final model detailed at the end of the section when corresponding with a grey box. The model creation and projection workflow consisted of 5 steps: 1) Allometric non-interacting model initialization, 2) Parameter tuning to data, 3) Turning on interactions within the model, 4) Simulating the status quo steady-state for 100 years to ensure numerical stability, 5) Simulating ssBH gradually implemented into the system for 100 years and then fishing the target ssBH regime for 700 years to ensure a steady-state is reached.

3.2. SPECIES SELECTION

All models were composed of fourteen commercially and ecologically important species with a variety of life histories (Table 1).

Table 1: Celtic Sea model species list and accompanying ecological information. Common length is taken from Fishbase, and biomasses estimate from Lauria (2016) with biomass weightings from Lauria (2012) for red gurnard and boarfish.

Common Name	Scientific name	Functional Role	Biomass (g/m ²)	Common Length (cm)
Mackerel	<i>Scomber scombrus</i>	Small pelagic predator/ forage fish	15.6530	30.0 FL
Horse mackerel	<i>Trachurus trachurus</i>	Small pelagic predator/ forage-like	8.2880	22.0 FL
Blue whiting	<i>Micromesistius poutassou</i>	Pelagic forage fish	2.0170	22.0 TL
Boarfish	<i>Capros aper</i>	Small pelagic/ benthopelagic predator	0.1502	13.0 SL
Monkfish	<i>Lophius piscatorius</i>	Demersal ambush predator	1.1740	100.0 SL
Whiting	<i>Merlangius merlangus</i>	Demersal Predator	0.9230	23.5 TL
Hake	<i>Merluccius merluccius</i>	Demersal predator	0.9050	45.0 TL
Atlantic cod	<i>Gadus morhua</i>	Demersal predator	0.7130	100.0 TL
Atlantic herring	<i>Clupea harengus</i>	Schooling forage fish	0.5640	30.0 SL
Megrim	<i>Lepidorhombus whiffiagonis</i>	Demersal predator/ benthic flatfish	0.3130	25.0 TL
Haddock	<i>Melanogrammus aeglefinus</i>	Benthivore/ omnivore	0.2588	35.0 TL
Plaice	<i>Pleuronectes platessa</i>	Flat fish, benthivore	0.1372	40.0 TL
Sole	<i>Solea solea</i>	Flatfish, benthivore	0.0580	35.0 SL
Red gurnard	<i>Chelidonichthys cuculus</i>	Demersal benthic predator/ benthivore	2.4735	27.6 FL

3.3. MODEL PARAMETERS

There were 35 species parameters for each of the 14 species included in the model and 4 gear parameters for each of the 5 gear types for each species. These are summarised, together with their sources in Table 2.

Table 2: Species and gear parameters included in the model. Red boxes refer to parameters which were tuned via data matching methods, those in blue were set to predetermined ecological estimates.

Parameter	Description	Sources/values
w_{max}	Maximum body mass (g)	Max reported size on Fishbase or within the landings data (European Commission, 2024).
w_{repro_max}	Asymptotic body mass (g)	Initially set equal to w_{max}
w_{mat}	Mass at maturity (g)	Initially set to glm fit to length-maturity data (DATRAS; Silva et al., 2024) or if data was not available set to averaged value from Fishbase.
age_{mat}	Age at maturity (years)	Set to averaged value from Fishbase
LWR (a, b)	Length-weight conversion	Set to averaged value from Fishbase
n	Allometric exponent for consumption/encounter	The exponent n is typically estimated between 2/3 and 3/4 (Kleiber 1932; West et al., 1997; Brown et al., 2004). Therefore, in this study the n of all species was set to 0.7.
p	Allometric exponent for metabolism	Equal to n (Andersen, 2019)
d	Allometric exponent for background mortality	Set to $n-1$ (Andersen, 2019)
q	Search volume allometric exponent	Set to 0.75 (Andersen, 2019)
w_{mat25}	Weight at 25% maturity (g)	Initially set to 50% of w_{mat} , or glm fit to length-maturity data (DATRAS, Silva et al., 2024); Tuned to all data fit types
E	Coefficient of external encounter	Tuned to all data fit types

$B_{observed}$	Biomass value of the group (g/km ²)	Lauria (2012, 2016)
B_{cutoff}	Size beyond which observed biomass values should be matched to model (g)	Set to w_{mat} if Lauria (2016) had a juvenile/adult split stanza for the species. Set to 1g if Lauria (2016) had a combined single stanza for the species.
μ_{mat}	Mortality at maturity size, set the coefficient for external mortality μ_i . (1/year)	Initially set to 0.2/year; Tuned to all data fit types
d/g	Proxy for diffusion coefficient D_i	Initially set to 0.15; Tuned to all data fit types
spawning_mu	Mean spawning date as fraction of a year in [0,1).	Datta and Blanchard (2016); Lucio et al. (1998); Alvarez et al. (2004); Alfonso-Dias and Hislop (1996); Corbin (2009); Seafish (2014); Farrell (2012); Egerton et al. (2017); Nichols et al. (1993)
spawning_kappa	Spawning concentration	Datta and Blanchard (2016); Lucio et al. (1998); Alvarez et al. (2004); Alfonso-Dias and Hislop (1996); Corbin (2009); Seafish (2014); Farrell (2012); Egerton et al. (2017); Nichols et al. (1993)
$annuli_{min\ age}$	Minimum age (years) at which first ring can form	Tuned to age at length data
$annuli_{date}$	Ring formation day as fraction of year [0,1).	Tuned to age at length data
$predation_{kernel}$	Predation kernel type	Set to “power law”
$kernel_{exp}$	The exponent of the power law	Set using estimates from MizerStomach fits
$kernel_{ll}$	The location of the left, rising sigmoid	Set using estimates from MizerStomach fits
$kernel_{ul}$	The shape of the left, rising sigmoid	Set using estimates from MizerStomach fits
$kernel_{lr}$	The location of the right, falling sigmoid	Set using estimates from MizerStomach fits
$kernel_{ur}$	The shape of the right, falling sigmoid	Set using estimates from MizerStomach fits
$alpha$	Assimilation efficiency	Set to 0.7 (Andersen, 2019)

k_s	Metabolic respiration coefficient	Initially set to 0, later set via back calculation (Section 8.1)
h	Maximum intake rate coefficient	Initially set to Inf, later set via back calculation (Section 8.1)
γ	Coefficient of the search volume	Internally estimated from allometric encounter scaling
m	Reproduction investment	Initially set to 1; Tuned to all data fit types
$R_{max}, erepro$	Reproductive efficiency and allocation	Internally estimated so populations maintain biomass (Section 8.3)
$interaction_{resource}$	Sets interaction strength with resource	Set to 0.
w_{min}	The minimum weight (g)	Set to 0.001 g
$Y_{observed}$	The observed tonnes of dead fish due to fishing (g/year/km ²)	Yields taken from Lauria (2012) and adjusted to include discards to get total tonnes of dead fish due to fishing (Section 9.3)
Q_g	Catchability of specific species and gear	Tuned to achieve observed yield
L_{50}	Length at which 50% of fish are caught (cm)	Tuned to survey and landings size distributions
L_{25}	Length at which 25% of fish are caught (cm)	Tuned to survey and landings size distributions
$gear$	Gear type	Set to trawl, pelagic, misc, gillnet, or survey.

3.4. MODEL FOUNDATIONS

3.4.1. POPULATION DYNAMICS

The *mizer* framework characterizes individuals by weight (w) and species index (i), calculating the size spectrum as the density of individuals across size classes $N_i(w)$. Individual level processes are scaled to the population using the McKendrick-von Foerster formulation (McKendrick, 1926; von Foerster, 1959; Silvert & Platt, 1978; Law et al., 2009; Scott et al., 2014), where density changes are governed by deterministic individual growth rates $g_i(w)$ and mortality rates $\mu_i(w)$, determined from three environmental drivers: food availability, predation and fishing mortality.

$$\frac{\partial N_i(w)}{\partial t} = -\frac{\partial g_i(w)N_i(w)}{\partial w} - \mu_i(w)N_i(w) \quad \text{Eq. 1}$$

To account for individual variability, this approach was extended by replacing deterministic growth with a with a stochastic formulation (Lv & Pitchford, 2007). In this formulation dynamics are governed by:

$$\frac{\partial N_i(w, t)}{\partial t} = -\frac{\partial J_i(w, t)}{\partial w} - \mu_i(w)N_i(w, t) \quad \text{Eq. 2}$$

Here the total flux $J_i(w, t)$ is composed of both advective and diffusive growth components (Datta et al., 2010, 2011):

$$J_i(w, t) = g_i(w)N_i(w, t) - \frac{1}{2} \frac{\partial}{\partial w} [D_i(w)N_i(w, t)] \quad \text{Eq. 3}$$

Substituting Equation 3 into Equation 2 makes the population dynamic equation below:

$$\frac{\partial N_i(w)}{\partial t} = -g_i(w)N_i(w, t) + \frac{1}{2} \frac{\partial}{\partial w} [D_i(w)N_i(w, t)] - \mu_i(w)N_i(w, t) \quad \text{Eq. 4}$$

This formulation is vital for matching parameters to population age-at-size data. As Lv and Pitchford (2007) demonstrate, the diffusion of population density causes the observed mean size to deviate from deterministic trajectories. This is compounded by size-dependent mortality, where lucky individuals that reach larger size classes more quickly experience reduced mortality, fundamentally shifting the mean biomass flow of the population. By incorporating this diffusive spread, the model accounts for the overestimation of growth parameters known as Lee's Phenomenon (1912), ensuring that production estimates for ssBH regimes are not skewed by the preferential survival of fast-growing individuals.

3.4.2. STEADY-STATE

The final results of this study are analysed at steady-state. Mathematically this occurs when the rate of change in population density over time is zero for every species (i) at every weight (w) in the model (Diekmann et al., 2003):

$$\frac{\partial N_i(w, t)}{\partial t} = 0 \quad \text{Eq. 5}$$

In this state, the system has reached a dynamic equilibrium where population densities across all size classes and species remain constant, despite the overall flux of individuals through the system. Reaching this equilibrium ensures that the results are due to underlying ecological interactions and fishing pressures, rather than transient numerical fluctuations from the model's initialization.

3.4.3. *MODEL PROJECTION*

Model projection involved simulating the parameterised model forward in time by iteratively solving the size-spectrum equation (Equation 4). At each time step, change in the number density was calculated and used to update abundance-at-size, producing a new number density for the next time point. Repeating this process allowed the system to evolve over time.

At every step, growth, mortality, and reproduction were recalculated from the current system state. Because these biological processes depend on current number density, changes in any single species or size class dynamically fed back across the entire system.

3.5. STATIC TO DYNAMIC STATES

The development of the Celtic Sea model was conducted in three distinct stages: the initialization of the allometric non-interacting model, parameter tuning of this model to match empirical data, and the enabling of interactions (Figure 1).

Within the allometric non-interacting model, species-specific background encounter and mortality rates were defined using power-law relationships. These rates represented the aggregate ecological pressures imposed by the wider ecosystem, allowing unresolved food web structure to be implicitly incorporated despite only 14 species being modelled explicitly.

When trophic interactions were subsequently enabled, the model retained its calibrated steady-state because predation and encounter events from explicitly modelled species emerged within the pre-established background mortality and encounter budgets, rather than introducing additive pressures.

3.6. NON-INTERACTING ALLOMETRIC MODEL

Individual-level processes of the population density equation (Equation 4) of the non-interacting model were parameterised using allometric scaling relationships. Within this framework, individual growth emerged from energetic processes governing prey encounter, metabolism and reproductive allocation.

Here the encounter rate $E_i(w)$ described the background frequency at which predator species (i) at weight (w) encountered potential prey, consistent with metabolic theory (Andersen, 2019):

$$E_i(w) = E_i w^n \quad \text{Eq. 6}$$

Where E_i was the coefficient of encounter and n was the exponent of encounter.

The net energy available for growth and reproduction ($E_{r,i}(w)$), was derived from this encountered energy after accounting for the feeding level ($f_i(w)$) (the proportion of encountered prey consumed), the assimilation efficiency (α_i), and energy lost to metabolism ($k_i(w)$).

$$E_{r,i}(w) = \alpha_i(1 - f_i(w))E_i(w) - k_i(w) \quad \text{Eq. 7}$$

In the model, metabolism and encounter scaled with the same exponent n :

$$k_i(w) = k_{s,i} w^n \quad \text{Eq. 8}$$

Where k_i was the coefficient of metabolic loss.

In the non-interacting model, both the feeding level and metabolic loss were set to 0 to allow the $E_{r,i}(w)$ to collapse into a single power law:

$$E_{r,i}(w) = E_{r,i} w^n \quad \text{Eq. 9}$$

Growth of individuals ($g_i(w)$) from Equation 4, was then formulated as the available energy not used for reproduction, where $\psi_i(w)$ represented the proportion allocated to reproduction:

$$g_i(w) = E_{r,i}(w)(1 - \psi_i(w)) \quad \text{Eq. 10}$$

To maintain mathematical consistency across the community density flux equation (Equation 4), background mortality and diffusion were scaled accordingly as:

$$\mu_i(w) = \mu_i w^{n-1} \quad \text{Eq. 11}$$

$$D_i(w) = D_i w^{n+1} \quad \text{Eq. 12}$$

Where μ_i was the mortality coefficient and D_i was the diffusion coefficient. By setting the mortality exponent to $n-1$ and the diffusion exponent to $n+1$, the model reconstructed the community scale-invariance empirically observed by Sheldon et al., (1972), ensuring that the empirical reality of the marine size spectrum was reflected (Capitán and Delius, 2010).

Model parameters from Table 2 were set to default or literature values to initialize the non-interacting model.

3.7. PARAMETER TUNING

After model initialization parameters were tuned to visually match data fits.

Total energy income was tuned via the encounter coefficient E_i .

Size-dependent growth patterns were tuned via parameters associated with the proportion of energy allocated to reproduction versus somatic growth including, maximum size (w_{repro_max}), maturity weight (w_{mat}, w_{mat25}), and the allocation exponent (m). These adjustments manipulated the allocation function $\psi_i(w)$:

$$\psi_i(w) = \left(1 + \left(\frac{w}{w_{mat,i}} \right)^{\frac{\log(3)}{\log\left(\frac{w_{mat,i}}{w_{mat25,i}}\right)}} \right)^{-1} \left(\frac{w}{w_{repro_max,i}} \right)^{m-n} \quad \text{Eq. 13}$$

The stochastic spread of growth was calibrated via d/g the proxy for the species-specific diffusion coefficient D_i (Equation 12; Table 2), and the total mortality was set as the sum of natural mortality (Equation 11) and the fishing mortality (Equation 15).

Natural background mortality was tuned via μ_{mat} a proxy for the mortality coefficient μ_i . Fishing mortality was tuned via the fishing parameters catchability (Q_{ig}) and selectivity parameters ($L_{50,i,g}$ and $L_{25,i,g}$).

Energetic balances were solved internally, and after each parameter adjustment the model was returned to a model at steady-state.

3.8. STEADY-STATE REPRODUCTION CALIBRATION

The tuned model was returned to a model at steady-state by balancing egg production against mortality losses. The conversion of reproductive energy to realized offspring production was modelled using a Beverton-Holt functional form (Scott et al., 2014), representing density-dependent limitation on the conversion of reproductive energy into surviving offspring.

The density-independent rate of egg production $R_{di,i}$, represented the total potential offspring production prior to density-dependence and was calculated by integrating the reproductive energy flux across the mature weight spectrum:

$$R_{di,i} = \frac{\epsilon_i}{2w_{min,i}} \int N_i(w) E_{r,i}(w) \psi_i(w) dw \quad \text{Eq. 14}$$

where ϵ_i was the reproductive efficiency and $w_{min,i}$ was the egg size.

The realized rate of offspring production $R_{dd,i}$ was set to exactly match mortality losses of the tuned state using the Beverton-Holt relationship:

$$R_{dd,i} = R_{di,i} \frac{R_{max,i}}{R_{di,i} + R_{max,i}} \quad \text{Eq. 15}$$

where $R_{max,i}$ was the maximum offspring production rate. During tuning, ϵ_i and $R_{max,i}$ were adjusted internally to ensure recruitment would maintain the tuned state abundance and biomass.

3.9. METHODS FOR DATA MATCHING

3.9.1. BIOMASSES

Observed biomass estimates were matched by rescaling the number density ($N_i(w)$) so that the sum of biomasses across all size classes for a species past the $biomass_{cutoff}$ matched an observed value.

Observed biomass ($grams/m^2$) estimates were obtained from Lauria (2016) Ecopath model of the Celtic Sea.

3.9.2. CATCHES AND YIELDS

Catch data, including size structure and total yields, were derived from both discards and landings. A post-discard mortality of 50% was assumed, resulting in status quo yields representing the dead fish biomass due to fishing (see Appendix). Survey-derived yields were assigned a nominal value of $10e^{-6}$ *grams/m²/year*, as total empirical yields from scientific surveys constitute a negligible fraction of total system mortality and are difficult to estimate robustly.

To align modelled catch with observed length-frequency distributions, gear selectivity for each species (i) and gear type (g), $S_{i,g}(l)$, was defined using a single sigmoid function:

$$S_{i,g}(l) = \frac{1}{1 + \exp\left(\log(3) \frac{l_{50,i,g} - l}{l_{50,i,g} - l_{25,i,g}}\right)} \quad \text{Eq. 16}$$

The selectivity function was evaluated over the length grid and converted to weight space using the inverse length-weight relationship (LWR):

$$L_i(w) = \left(\frac{w}{a_i}\right)^{\frac{1}{b_i}} \quad \text{Eq. 17}$$

yielding:

$$S_{i,g}(w) = S_{i,g}\left(\left(\frac{w}{a_i}\right)^{\frac{1}{b_i}}\right) \quad \text{Eq. 18}$$

Status quo fishing mortality was then defined as:

$$F_{i,g}(w) = S_{i,g}(w)Q_{i,g} \quad \text{Eq. 19}$$

Certain gears, such as gillnets, are often characterized by double-sigmoid selectivity functions (Kuparinen, et al., 2009). However, a single sigmoid function was adopted for all gear types here due to the aggregation of multiple mesh sizes and target assemblages within each gear category. This aggregation produced a composite selectivity pattern that is well approximated by a sigmoidal curve, while also improving model parsimony and computational efficiency.

Selectivity was subsequently combined with the modelled population number densities ($N_i(w)$) to obtain normalized modelled catches ($C_{i,g}(w)$).

$$C_{i,g}(w) = \frac{S_{i,g}(w)N_i(w)}{\int S_{i,g}(w)N_i(w) dw} \quad \text{Eq. 20}$$

The resulting catch densities were transformed back into length space using a Jacobian correction to ensure consistency with observed survey and landing length distributions:

$$C_{i,g}(l) = C_{i,g}(a_i l^{b_i}) * a_i b_i l^{b_i-1} \quad \text{Eq. 21}$$

Model normalized catches and observed normalized catches were plotted and parameters were tuned iteratively through visual comparison to ensure alignment between modelled and observed catch length-frequency distributions.

Total modelled yield for each species and gear ($Y_{i,g}$) was obtained by integrating across all size classes, where yield is defined as the product of selectivity, catchability, population density, and the weight:

$$Y_{i,g} = \int S_{i,g}(w) Q_{i,g} N_i(w) w dw \quad \text{Eq. 22}$$

Parameters were adjusted until total modelled yields closely matched reconstructed observed yields, ensuring the model accurately represented the current exploitation state of the Celtic Sea.

3.9.3. AGE AT LENGTH DISTRIBUTIONS

Model predictions were matched to observed age-length data to tune parameters governing growth, mortality, and reproduction, so that the model reproduced the observed population age-length structure derived from survey data (see Section 9.2.3 and 9.2.4).

To match length-stratified empirical datasets collected across multiple survey dates, the approach involved: convolving a single-cohort simulation with an annual spawning distribution, mapping modelled ages to annuli, predicting the age distribution at length for each survey date, and evaluating model fit using a likelihood-based framework.

First, the population density equation (Equation 2) was used to project an initial impulse of recruits at the minimum size (w_{min}) forward to an age exceeding the maximum observed annuli count by two years. This produced the size-age distribution $G_i(w, a)$, representing the proportion of an initial cohort of fish born at the same instant that survives to age a and reaches size w . This distribution was then rebinned from model weight bins into length classes corresponding to those used in the survey data, yielding $G_i(l, a)$.

Next, this size-age distribution was weighted by the species-specific spawning intensity, $S_i(\tau(t))$, to give the expected population structure at survey date t_{survey} :

$$N_{pop,i}(l, a|t_{survey}) = G_i(l, a) * S_i(\tau(t_{survey} - a)) \quad \text{Eq. 23}$$

where $t_{survey} - a$ represents the birth date of individuals of age a .

Spawning intensity, $S_i(\tau(t))$, was modelled using a von Mises distribution parameterised by the spawning time ($spawning_{\mu,i}$) and concentration values ($spawning_{\kappa,i}$) from literature (Table 2). Here, $\tau(t)$ denotes the fractional time of year corresponding to calendar date t .

A key challenge in comparing model predictions to observations is that observed ages (annuli counts) are not direct measures of chronological age but instead reflect the number of completed seasonal growth cycles. Because annuli formation depends on seasonal variation in growth, individuals of identical true age may exhibit different annuli counts depending on their birth timing relative to seasonal cycles (Chen et al., 2022). Additionally, even when an individual has experienced a full seasonal cycle, the most recently formed annulus may not yet be detectable as sufficient subsequent growth is required for the ring to become clearly identifiable.

To account for this, the model explicitly distinguished between true age (a) and observed age (K). Continuous ages (a) were mapped to annuli counts (K) using a deterministic annulus formation model defined by the timing of ring formulation and a minimum age threshold for first ring deposition (Spence and Turtle, 2017). These were treated as tuneable parameters. The true age-length population structure was then aggregated according to this age-ring mapping to obtain the expected number of fish of length l that would be observed with K annuli at survey time t_{survey} , $N_{model}(l, K|t_{survey})$.

From this, the probability of observing K annuli for a fish of length l at survey date t_{survey} was:

$$P(K|l, t_{survey}) = \frac{N_{model}(l, K|t_{survey})}{\sum_{K'} N_{model}(l, K'|t_{survey})} \quad \text{Eq. 24}$$

Observed data were represented as counts $C(K|l, j)$, denoting the number of fish of length l with K annuli in survey j . For each survey and length class, these counts were assumed to arise from a multinomial distribution with probabilities $P(K|l, t_{survey}, j)$.

The total negative log-likelihood (NLL) across all surveys, length classes, and annuli counts was calculated as:

$$NLL = - \sum_j \sum_l \sum_k C_{(K=k|l,j)} \log (P(K = k|l, t_{survey}, j)). \quad \text{Eq. 25}$$

The likelihood was used as a diagnostic metric to guide model parameter tuning. Parameters were iteratively adjusted to minimise the NLL and reduce systematic discrepancies between predicted and observed age-length distributions.

3.10. MATCHED

To achieve the non-interacting matched allometric Celtic Sea status quo model, the listed model parameters were tuned to visually maximize the fits to all data types simultaneously using a shiny app which had sliders for tuneable parameters and corresponding age likelihood and catch plots.

3.11. TURNING ON INTERACTIONS

Species interactions were enabled by incorporating dynamic density-dependent biological processes of the modelled species into the tuned allometric rates. This involved parameterising feeding levels, incorporating predation into both encounter and mortality rates, and implementing density-dependent reproduction.

3.11.1. FEEDING LEVELS

Feeding rates were assumed to follow a Holling type II functional response curve, where the feeding level ($f_i(w)$) was defined as the proportion of the maximum intake rate ($h_i(w)$) at which a predator i consumed encountered prey ($E_i(w)$):

$$f_i(w) = \frac{E_i(w)}{E_i(w) + h_i(w)} \quad \text{Eq. 26}$$

Following, Andersen (2019), the feeding level of stable marine systems was assumed to be approximately 0.6 and the feeding level at which the metabolic losses of a species exactly balances the ingested energy, such that growth and reproduction cease, also known as the critical feeding level (level ($f_{c,i}(w)$)) was assumed to approximate 0.2. These recommended values were used to supply realistic regimes and avoid persistent starvation or satiation within the model.

To incorporate these constraints into the tuned non-interacting model, the encounter rate coefficient (E_i), maximum intake rate coefficient (h_i), and metabolic loss coefficient ($k_{s,i}$), were back-calculated from the target feeding levels and the available energy for growth and reproduction $E_{r,i}(w)$. This was achieved by combining the energy balance equation (Equation 7) with the feeding level calculation (Equation 26) and enforcing the condition that net energy is zero at the critical feeding level.

$$E_i = E_{r,i} \frac{f_i}{\alpha_i(f_i - f_{c,i})(1 - f_i)} \quad \text{Eq. 27}$$

$$h_i = E_{r,i} \frac{1}{\alpha_i(f_i - f_{c,i})} \quad \text{Eq. 28}$$

$$k_{s,i} = E_{r,i} \frac{f_{c,i}}{f_i - f_{c,i}} \quad \text{Eq. 29}$$

3.11.2. PREDATION

Introducing predation reformulated encounter and mortality rates from static power-law representations of system-wide processes into dynamic, interaction-driven functions mediated by species composition and size structure, embedded within the underlying tuned background system.

Predation was parameterised using size-dependent search volumes ($\gamma_i(w)$), number densities ($N_i(w)$), feeding kernels ($\varphi_i\left(\frac{w}{w'}\right)$), and a species interaction matrix (θ_{ij}). Search volume followed $\gamma_i w^{0.75}$ (Andersen, 2019), with species-specific coefficients (γ_i) inherited from the initial allometric model creation. Number densities were taken directly from the matched baseline model. Predation kernels defined predator size selectivity and were fitted to empirical stomach content data (see Appendix). The interaction matrix was derived by dividing the target prey specific consumption rates, obtained from the diet matrix of Lauria (2016), by the corresponding baseline encounter rates. This scaling ensured that predator-prey interactions reproduced observed diet compositions given prey availability and size dependent selectivity (see Appendix).

The inclusion of predation in the encounter term resulted in the following formulation:

$$E_i(w) = \gamma_i(w) \int \left(\sum_j \theta_{ij} N_j(w') \right) \varphi_i\left(\frac{w}{w'}\right) w' dw' + E_{ext,i}(w) \quad \text{Eq. 30}$$

where $E_i(w)$ described the rate at which a predator species i of body size w acquired energy through feeding. The first term represents encounter rates arising from explicitly modelled species' predator-prey interactions, where prey availability was determined by prey species j abundances $N_j(w')$, interaction strengths between the species (θ_{ij}), and the size-based feeding selectivity ($\varphi_i\left(\frac{w}{w'}\right)$), scaled by size-dependent search efficiency ($\gamma_i(w)$). The second term represents the residual background ecosystem encounter not resolved by the interaction network. This external encounter term ($E_{ext.i}(w)$) represented the remaining proportion of the tuned power-law encounter $E_i w^n$ after subtracting the explicitly modelled predation encounters, following the formulation:

$$E_{ext.i}(w) = \max\left(0, E_i w^n - \gamma_i(w) \int \left(\sum_j \theta_{ij} N_j(w')\right) \varphi_i\left(\frac{w}{w'}\right) w' dw'\right) \quad \text{Eq. 31}$$

This ensured that the total encounter accurately reflected both fixed background intake from the system and dynamic consumption via predation within the modelled community.

The inclusion of predation in the mortality rate was analogously formulated:

$$\mu_i(w) = \sum_j \theta_{ji} \int \gamma_j(w') \varphi_j\left(\frac{w'}{w}\right) (1 - f_j(w')) N_j(w') dw' + F_i(w) + \mu_{ext.i}(w) \quad \text{Eq. 32}$$

where $\mu_i(w)$ described the total mortality rate of species i of body size w . The first term represented predation mortality on species i , determined by predator species j abundances ($N_j(w')$), search efficiency ($\gamma_j(w')$), feeding kernels ($\varphi_j\left(\frac{w'}{w}\right)$), interaction strengths with species i (θ_{ji}), and realised feeding levels ($1 - f_j(w')$). The second term represented the fishing mortality $F_i(w)$ from Equation 19 summed across all gears.

The third term represented the residual background mortality not resolved by the interaction network. This external mortality term ($\mu_{ext.i}(w)$) represented the remaining proportion of the tuned power-law mortality $\mu_i w^n$ after subtracting the explicitly modelled predation mortality, following the formulation:

$$\mu_{ext.i}(w) = \max\left(0, \mu_i w^n - \sum_j \theta_{ji} \int \gamma_j(w') \varphi_j\left(\frac{w'}{w}\right) (1 - f_j(w')) N_j(w') dw'\right) \quad \text{Eq. 33}$$

This ensured that the total mortality accurately reflected both fixed background losses and dynamic predation pressure within the modelled community.

This extension preserved the tuned steady-state structure of the Celtic Sea matched status quo model while embedding explicit trophic interactions within it, as predation rates did not extend beyond the matched power laws. For future model projections, these changes allowed species interactions responded dynamically to changes in the community composition and fishing pressure, at times extending beyond the background power laws.

3.11.3. REPRODUCTION

With the steady-state established and predation interactions enabled, the model's sensitivity to external perturbations (i.e. shifts in fishing pressure) was parameterized via the reproduction level.

While multiple combination of $R_{max,i}$ and ϵ can yield the same steady-state $R_{dd,i}$, the ratio between the $R_{max,i}$ and the $R_{dd,i}$ determines the steepness of the recruitment curve, and thus the system's resilience. The reproduction level is defined as the ratio between the density-dependent recruitment and the maximum recruitment capacity:

$$Reproduction\ Level_i = \frac{R_{dd,i}}{R_{max,i}} \quad \text{Eq. 34}$$

In this study, the reproduction level was utilized to tune the resilience of the system. The value chosen determines how a species responds to perturbations. Although this parameter can be derived from Maximum Sustainable Yield (MSY) curves, it was fixed at 0.8 for all species in this simulation. This value was selected to prioritize species resilience, ensuring the model remained robust under the projected fishing regimes.

This step marked the completion of the interacting Celtic Sea model.

3.12. CALCULATING SSBH FISHING

The ssBH fishing rule defined by Law and Plank (2022) is:

$$Y_i(w) \propto P_i(w)B_i(w) \quad \text{Eq. 35}$$

or

$$Y_i(w) = c P_i(w)B_i(w) \quad \text{Eq. 36}$$

where for each species i of size w , $Y_i(w)$ is the yield ($g/m^2/year$), $P_i(w)$ is the production ($g/m^2/year$), $B_i(w)$ is the biomass (g/m^2), and c is the proportionality

constant (m^2/g) which can be thought of as a fishing intensity of the balanced harvest regime.

Since yield is defined as the fishing mortality $F_i(w)$ (1/year) multiplied by the biomass:

$$Y_i(w) = F_i(w)B_i(w) \quad \text{Eq. 37}$$

Equation 36 can be substituted into Equation 37 to give:

$$F_i(w) = c P_i(w) \quad \text{Eq. 38}$$

Species production at each time step t ($P_i(w, t)$) was calculated within the model at each time step by taking the instantaneous flux of individuals moving through the size w ($J_i(w, t)$) from Equation 3, and multiplying it by the corresponding weight:

$$P_i(w, t) = w J_i(w, t) \quad \text{Eq. 39}$$

To model ssBH fishing mortality the adjusted ssBH rule from Equation 38 was substituted into Equation 39.

$$F_{ssBH,i}(w, t) = c w J_i(w, t) \quad \text{Eq. 40}$$

Essentially, under the balanced harvest regime, fishing mortality was the proportionality constant multiplied by weight and the flux.

3.13. PROJECTING SSBH

3.13.1. OVERVIEW

SsBH mortalities could not be applied directly to the calibrated interacting model, as the ssBH regime represented a major departure from the current ecosystem. To allow a gradual transition to the ssBH regime. The model status-quo model was projected forward, with the fishing mortality array set as a weighted combination of status-quo and ssBH fishing, slowly increasing towards the desired weighting of ssBH until the desired weighting was reached and the model was projected until it reached steady-state. This allowed progressive adjustments to the new fishing regime increasing model stability.

A total of 220 ssBH models were made with different combinations of balanced harvest fishing intensities(c) and proportion of fishing from ssBH and status-quo fishing (A). The

c values tested included values ranging from 0 to 1 with increments of one tenth and integers between 1 to 10. The A values tested included values ranging from 0 to 1 with increments of one tenth.

3.13.2. GRADUAL IMPLEMENTATION

The combined fishing mortality at each time step of model projection was calculated by the following formulation. Where $F_{Status,i}(w, t)$ is given by the sum across all gears in Equation 19 and $F_{PssBH,i}(w, t)$ is given:

$$F_{combined,i}(w, t) = (1 - A)F_{Status,i}(w, t) + AF_{PssBH,i}(w, t) \quad \text{Eq. 41}$$

Where A was the proportion of fishing mortality allocated from ssBH. For the initial 100 years ($t_{steadied} = 100 \text{ years}$) of projection the model was run with the alpha set to 0, where the status quo fishing was implemented to ensure an absolute steady-state of the Celtic Sea model was reached. For the next 100 years ($t_{blended} = 200 \text{ years}$) of model projection A was slowly increased at each time step via the following equation:

$$A = A_{max} \frac{(t - t_{steadied})}{t_{blended} - t_{steadied}} \quad \text{Eq. 42}$$

Where t was the current year of the model projection step and A_{max} was the desired final weighting of ssBH in the model. After A_{max} was reached ($t = t_{blended}$) the A value in Equation 40 was simply set to A_{max} and the model was projected forward for another 500 years ($t_{duration} = 700 \text{ years}$) to ensure steady-state conditions were met. This was repeated for each unique combination of A and c values.

3.14. EVALUATION METRICS

Ecosystem state was evaluated using the biological metrics: number of individuals (N), species biomass (B), spawning stock biomass (SSB), yield (Y), total biomass and proportion biomass of large fish in yield (BLFY and PBLFY), and total biomass and proportion biomass of large fish in the system (BLFS and PBLFS). Large fish were defined as all individuals larger than 50cm as recommended by Shephard et al., 2011.

The abundance at time step t was calculated as:

$$N_i(t) = \int N_i(w, t) dw \quad \text{Eq. 43}$$

for species (i), and as

$$N_{total}(t) = \sum_i \int N_i(w, t) dw \quad \text{Eq. 44}$$

for the total abundance in the system.

The biomass at time step t was calculated as:

$$B_i(t) = \int N_i(w, t)w dw \quad \text{Eq. 45}$$

for species (i), and as

$$B_{total}(t) = \sum_i \int N_i(w, t)w dw \quad \text{Eq. 46}$$

for the total biomass in the system.

SSB was the biomass of all individuals past maturity size and was calculated for time step t as:

$$SSB_i(t) = \int_{w_{mat}}^{w_{max}} N_i(w, t)w dw \quad \text{Eq. 47}$$

for species (i), and as

$$SSB_{total}(t) = \sum_i \int_{w_{mat}}^{w_{max}} N_i(w, t)w dw \quad \text{Eq. 48}$$

for the total abundance in the system.

Total ecosystem yield at time t for a given BH scenario was:

$$Y_i(t) = \int F_{combined,i}(w, t)B_i(w, t)dw \quad \text{Eq. 49}$$

for species (i), and as

$$Y_{total}(t) = \sum_i \int F_{combined,i}(w, t)B_i(w, t)dw \quad \text{Eq. 50}$$

for the total yield.

Large fish metrics were only calculated summed for all species; individual species values were not separately saved.

Total biomass fish above 50 cm in catches was evaluated as

$$BLFY = \sum_i \int_{w=a_i 50^{b_i}}^{w_{max}} F_{combined,i}(w, t) N_i(w, t) dw \quad \text{Eq. 51}$$

Proportion fish above 50 cm in catches was evaluated as:

$$PBLFY = \frac{\sum_i \int_{w=a_i 50^{b_i}}^{w_{max}} F_{combined,i}(w, t) N_i(w, t) dw}{\sum_i \int F_{combined,i}(w, t) N_i(w, t) dw} \quad \text{Eq. 52}$$

Total biomass fish above 50 cm in the system was evaluated as:

$$BLFS = \sum_i \int_{w=a_i 50^{b_i}}^{w_{max}} N_i(w, t) dw \quad \text{Eq. 53}$$

Proportion fish above 50 cm in the system was evaluated as:

$$PBLFS = \frac{\sum_i \int_{w=a_i 50^{b_i}}^{w_{max}} N_i(w, t) dw}{\sum_i \int N_i(w, t) dw} \quad \text{Eq. 54}$$

The Status quo Celtic Sea Model steady-state metric was evaluated from any BH simulation at $t = 100 \text{ years}$, the metric of each simulation of ssBH was evaluated at $t = 700 \text{ years}$.

To identify the optimal harvest strategy across all simulations, a multi-criteria composite score (MCCS) was calculated using the steady-state metric results of each model. Each metric was first normalized using min-max scaling (range normalization) to a 0-1 scale, where 1 represented the best observed performance and 0 the worst. Metrics were categorized into two primary indices: Conservation and Economic. The Conservation index was calculated as the average of the total N, B, SSB, BLFS and PBLFS. The Economic index was calculated as the average of total Y, BLFY, and PBLFY. The final performance score for each model was determined by the sum of these two indices, providing an equal weighting between ecological sustainability and fishery output.

3.15. SENSITIVITY ANALYSIS

A Morris sensitivity analysis was performed on key model parameters, including the background encounter rate coefficient (E_i), background mortality coefficient (μ_i),

diffusion coefficient (D_i), maturation size (l_{mat}), the mass-length scaling exponent (m), and maximum body size (w_{max}). Each parameter was perturbed by $\pm 10\%$ for each of the 14 species. Due to the high computational cost associated with exploring a large parameter space, the analysis was restricted to this subset of parameters.

These parameters were selected because their values were strongly influenced by model tuning processes and were, therefore, subject to greater uncertainty. In addition, they directly affected core model processes including feeding, mortality, growth, and reproduction. These parameters were also identified as highly influential in previous work (Benoit et al.), providing further justification for their inclusion in the sensitivity analysis.

The top 8 most influential parameters identified from the Morris test were then varied by 10% in a Latin hyper cube sampling and assessed via regional sensitivity analysis (RSA). RSA analysis revealed if parameter perturbations resulted in different conclusions in metrics system effects. RSA was computed on the steady-state of status quo, whole-ssBH fishing regimes of fishing intensity 0.1, 0.2, 0.3, 0.4, 1, and 10, as well as the hybrid regime with fishing intensity 0.4 and weighting 0.5. Status quo was run for 500 years till status quo steady-state was reached at $t = 500$, then blended to BH for 100 years then run for another 500 years till the respective BH regime steady-state was reached at $t = 1100$.

4. RESULTS

Here the various schemes of exploitation applied to the system were examined. Results were calculated at the steady-state under the prevailing pattern of fishing mortality.

Fishing regimes are separated by two primary variables: the weighting variable, A , which represents the proportion of fishing mortality originating from the ssBH regime versus the status quo gear-based methods, and the ssBH fishing intensity variable, c , which is the proportionality constant scaling the fishing mortality to production.

From these variables three distinct categories of exploitation were defined: status quo ($A = 0$), hybrid ($c > 0$ and $0 < A < 1$), and whole-ssBH ($c > 0$ and $A = 1$). The latter two are both included under the term “ssBH regime”.

Model species and gear parameters after tuning to data fit are recorded in the appendix (Section 7.3).

4.1. EFFECTS OF FISHING INTENSITY (c) IN WHOLE-SSBH

Investigation into whole-ssBH regimes ($A = 1$) revealed distinct response patterns across ecological metrics as fishing intensity (c) increased (Figure 2 and 3).

Total Y , B , SSB and N were highest at low fishing intensities ($c \leq 0.4$), with B and SSB exhibiting monotonic, exponential declines as c increased, while Y and N peaked at low intensities, $c = 0.4$ and $c = 0.2$, respectively, before declining (Figure 2a & 2d).

These aggregate declines were primarily driven by the depletion of species which at low intensities were most abundant: horse mackerel, mackerel, blue whiting, and boarfish. Conversely, species that were least abundant at low intensities, such as herring, red gurnard, and plaice, increased in Y , B , and SSB at higher fishing intensities (Figure 3). While these species-level increases were of a lower magnitude than the declines of dominant species and thus were not observable in the total system metrics, they indicate that higher fishing intensities promote a homogenization of biomass and yield across the community. Notably, this phenomenon did not extend to species abundances, as most less-abundant species' N remained relatively constant or decreased at higher intensities.

The PBLFS increased with fishing intensity, reaching an asymptote at $c \approx 3$, while the BLFS decreased across the entire range of c (Figure 2e & 2f). This confirms that the rise in PBLFS is a mathematical artifact of the rapid depletion of small, abundant pelagic species, rather than a genuine increase in large fish biomass.

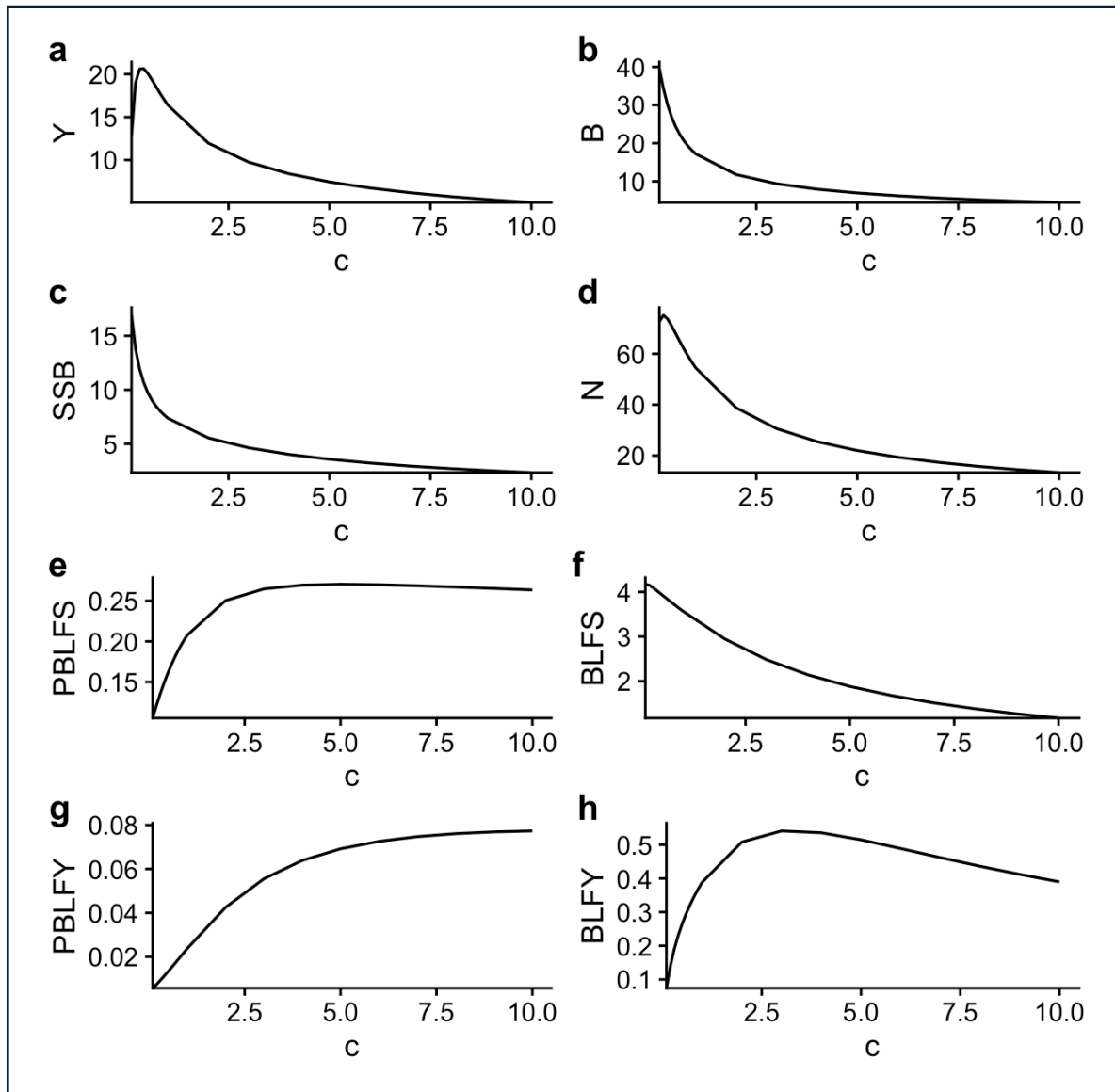


Figure 2: Steady-state system total evaluation metrics of whole-ssBH regimes with various fishing intensities (c). Evaluation metrics include: a) total yield ($g/year/m^2$), b) total biomass (g/m^2), c) total spawning stock biomass (g/m^2), d) number of individuals ($1/m^2$), e) proportion of biomass of large fish in the system, f) total biomass of large fish in the system (g/m^2), g) proportion of biomass of large fish in the yield, h) total biomass of large fish in the yield ($g/m^2/year$).

The BLFY peaked at $c = 3$ before declining (Figure 2h). This initial rise was driven by increased yields of larger-bodied species, including whiting, haddock and monkfish. Beyond this point, however, yields for cod, haddock and megrim declined, causing total BLFY to fall. The PBLFY increased with c across all intensities (Figure 2g), though the slope flattened at high intensities. This continued rise in PBLFY occurred because while both small pelagic yields and BLFY declined, the former decrease more rapidly, allowing the ratio to increase despite the falling harvest of large fish.

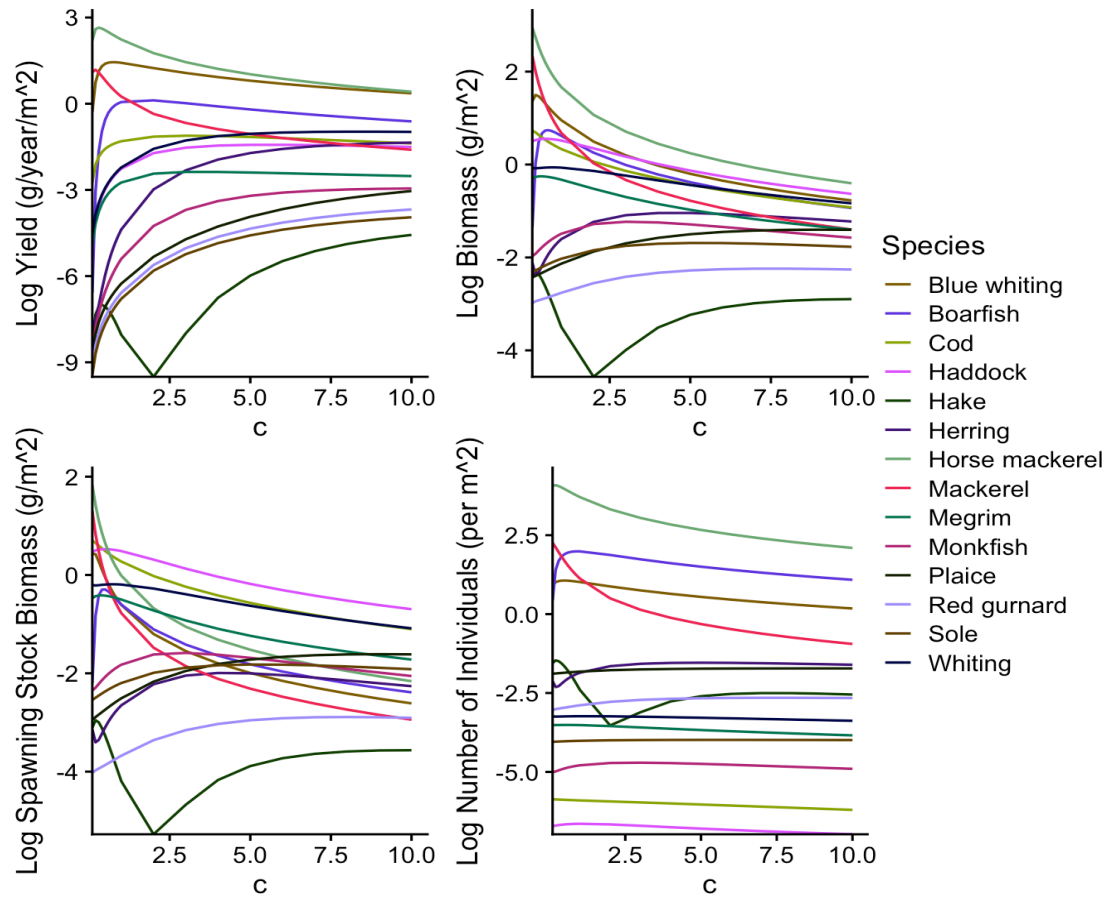


Figure 3: Steady-state species specific values for system evaluation metrics of whole-ssBH regimes with various fishing intensities. Evaluation metrics include yield, biomass, spawning stock biomass, and number of individuals. Metrics are logged to allow for visual resolution of less abundant species.

4.2. COMPARING TOTAL METRIC STATUS QUO TO SSBH REGIMES

The status quo model steady-state served as the baseline for all comparative analyses. Heatmaps across the (A, c) parameter space reveal the performance of alternative regimes relative to the status quo (Figure 4). Status quo metrics were $Y = 3.797 \text{ g/m}^2/\text{year}$; $B = 33.15 \text{ g/m}^2$; $SSB = 14.79 \text{ g/m}^2$; $N = 63.12/\text{m}^2$; $PBLFY = 0.22$, $BLFY = 0.83 \text{ g/m}^2/\text{year}$; $PBLFS = 0.03$; $BLFS = 1.09 \text{ g/m}^2$.

4.2.1. YIELD, BIOMASS, SSB, N

Across all steady-state hybrid and whole-ssBH regimes, yields were higher than the status quo. The absolute highest yield was $22.90 \text{ g/m}^2/\text{year}$ at $c = 4$ and $A = 0.1$. For whole-ssBH simulations, the maximum yield was $20.65 \text{ g/m}^2/\text{year}$ at $c = 0.4$.

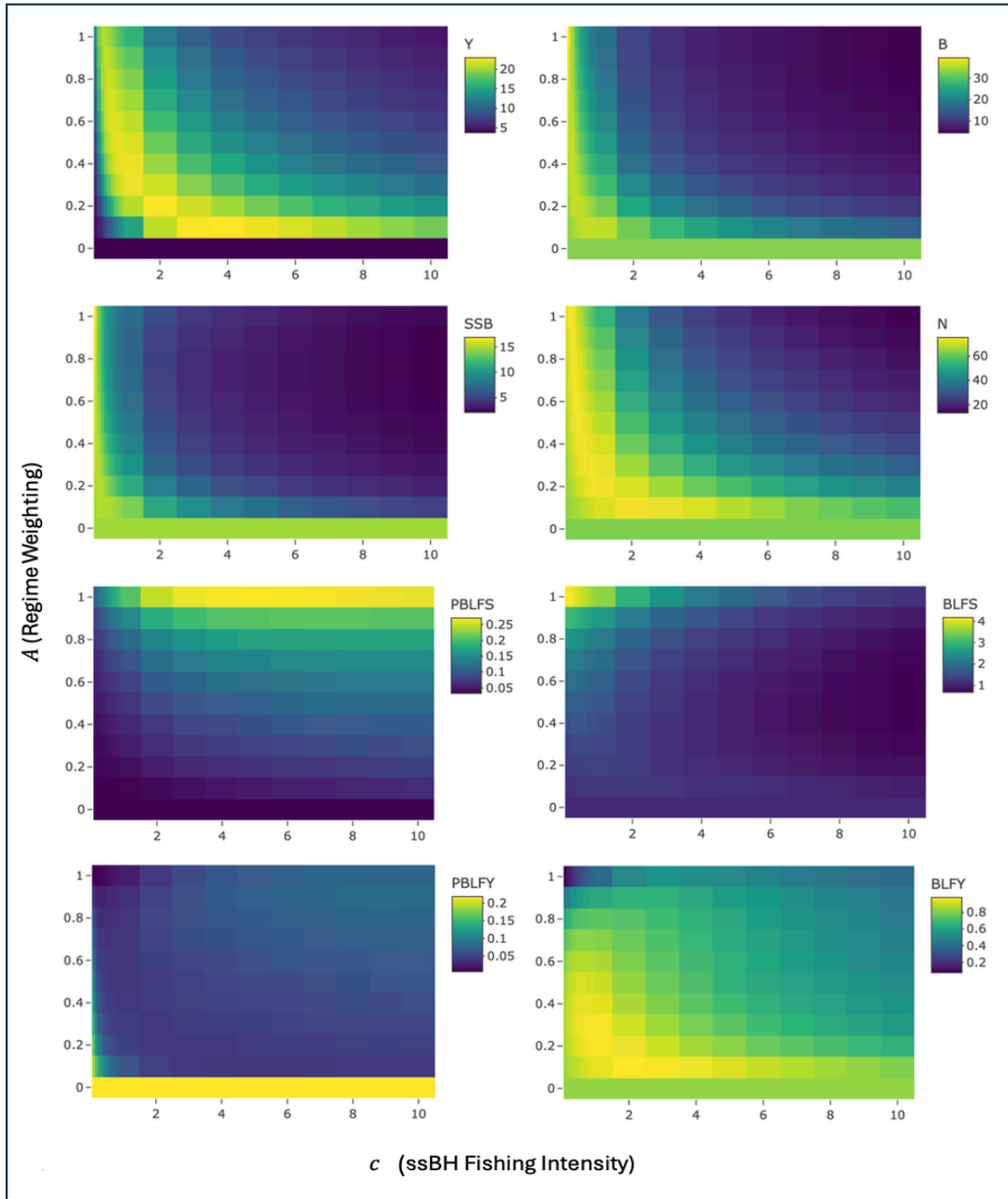


Figure 4: Heat map of steady-state values for system evaluation metrics of all fishing regimes. Status quo metric is the same for all $A = 0$, independent of the ssBH Fishing intensity. $A = 1$ represents the whole-ssBH fishing regimes with various fishing intensities as indicated on the x-axis. All other values plotted are hybrid regimes where ssBH fishing regimes is blended with the status quo proportionally, that is at $A = 0.2$, the hybrid regime is composed of 80% gear based-fishing and 20% ssBH regime with the corresponding fishing intensity. Metrics include yield ($g/m^2/year$), biomass (g/m^2), spawning stock biomass (g/m^2), number of individuals ($1/m^2$), proportion of large fish in system, biomass of large fish in system (g/m^2), proportion of biomass of large fish in yield, and biomass of large fish in yield ($g/m^2/year$).

For Biomass, SSB and N, the superiority of BH over the status quo was defined by specific power-law relationships between A and c .

- Biomass: $A < 0.093c^{-1.53}$
- SSB: $A < 0.026c^{-1.62}$
- N: $A < 0.61c^{-1.15}$

BH regimes falling above these power-laws resulted in equal or lower values compared to the status quo.

The absolute highest biomass was 39.54 g/m^2 at $c = 0.1$ and $A = 1$. The absolute highest SSB was 18.24 g/m^2 for the steady-state of the regime defined by $c = 0$ and $A = 1$ (no fishing scenario). For whole-ssBH simulations, the maximum SSB was 16.90 g/m^2 at $c = 0.1$.

The absolute highest N was 75.14 g/m^2 at $c = 0.2$ and $A = 1$.

4.2.2. LARGE FISH IN YIELD

Across all steady-state hybrid and whole-ssBH regimes, the proportion of large fish in yield were lower than the status quo. The highest ssBH regime PBLFY was 0.19 at $c = 0.1$ and $A = 0.1$. For whole-ssBH simulations, the maximum PBLFY was 0.08 at $c = 10$.

For BLFY, the superiority of BH over the status quo was variable amongst combinations of A and c , and the relationship was not easily defined by a power-law. The absolute highest BLFY was $0.98 \text{ g/m}^2/\text{year}$ at $c = 0.2$ and $A = 0.1$. No whole-ssBH simulation outcompeted the status quo, with the highest whole-ssBH BLFY $0.54 \text{ g/m}^2/\text{year}$ at $c = 3$.

4.2.3. LARGE FISH IN SYSTEM

Across all steady-state hybrid and whole-ssBH regimes, the proportion of large fish in the system was higher than the status quo. The highest BH regime PBLFS was 0.27 at $c = 5$ and $A = 1$.

For BLFS, the superiority of BH over the status quo was variable amongst combinations of A and c , and the relationship was not easily defined by a power-law. The absolute highest BLFS was 4.17 g/m^2 at $c = 0.1$ and $A = 1$.

4.3. OPTIMIZING METRICS

While whole-ssBH regimes had higher metric values than hybrid and status regimes in 4 of the 8 system evaluation metrics above, the MCCS identified hybrid management regimes as the top performing states. The top ten scoring regimes featured fishing intensities between 0.3 and 0.8 and weightings between 0.2 and 0.6. The worst scoring regimes had high weightings and high fishing intensities. With the top ten worst scoring regimes having a fishing intensity of 9-10 and weightings of 0.5 to 1.

The overall best-performing model was identified at the hybrid regime $c = 0.4$ and $A = 0.5$. The best whole-ssBH regime was $c = 0.4$. From this, three key regimes were identified: the hybrid ($c = 0.4$ and $A = 0.5$), whole-ssBH ($c = 0.4$ and $A = 1$), and status quo ($A = 0$). The steady-state of each of these regimes will from this point forward be referred to as such (Table 3).

Across the three key models the ssBH regimes had yields more than 5 times greater than the status quo. Biomass of the hybrid and status quo regime were comparable while the whole-ssBH regime had much lower biomasses. The spawning stock biomass was greatest in the status quo model. Abundance was greater in the ssBH regimes. Hybrid had larger BLFY than any model, and greater amounts of large fish in the system.

Table 3: Evaluation metric values for the status quo ($A = 0$), best whole-ssBH ($c = 0.4$), and best overall regime (hybrid)($c = 0.4$ and $A = 0.5$).

Evaluation Metric	Status Quo	Whole BH	Hybrid
Yield ($g/m^2/year$)	3.80	20.65	19.99
Biomass (g/m^2)	33.15	26.92	32.45
SSB (g/m^2)	14.79	10.67	11.95
N ($1/m^2$)	63.12	71.09	74.21
PBLFS	0.03	0.15	0.05
BLFS (g/m^2)	1.09	4.01	1.77
PBLFY	0.22	0.01	0.05
BLFY ($g/m^2/year$)	0.83	0.23	0.92
Conservation MCCS	0.51	0.71	0.56
Economic MCCS	0.47	0.39	0.63
MCCS	0.98	1.10	1.19

4.4. KEY STATES

The optimal BH regimes induced a fundamental restructuring of community composition and harvest profiles (Figure 6 & 7). Species specific yield and biomass values are contained within Table 4.

4.4.1. SPECIES AND SIZE-SELECTIVE FISHING MORTALITY AND YIELD

The comparison of the three key regimes reveals fundamental differences in harvest profiles and community biomass equilibrium. The status quo steady-state was characterized by high fishing mortality concentrated on the largest size classes, typically between 100 g-10,000 g, and primarily target demersal species. Consequently, yields in this state were focused on individuals above 50g and were distributed with relative evenness across species, with the highest yields in order from mackerel, whiting, cod, and horse mackerel (Table 4 and Figure 5).

In contrast, the steady-state for the whole-ssBH regime ($c=0.4$) was defined by a mortality pressure shifted almost exclusively toward the smallest size classes, ranging from 0.1 g to 100 g, while fishing mortalities of the status quo were sigmoidal, the

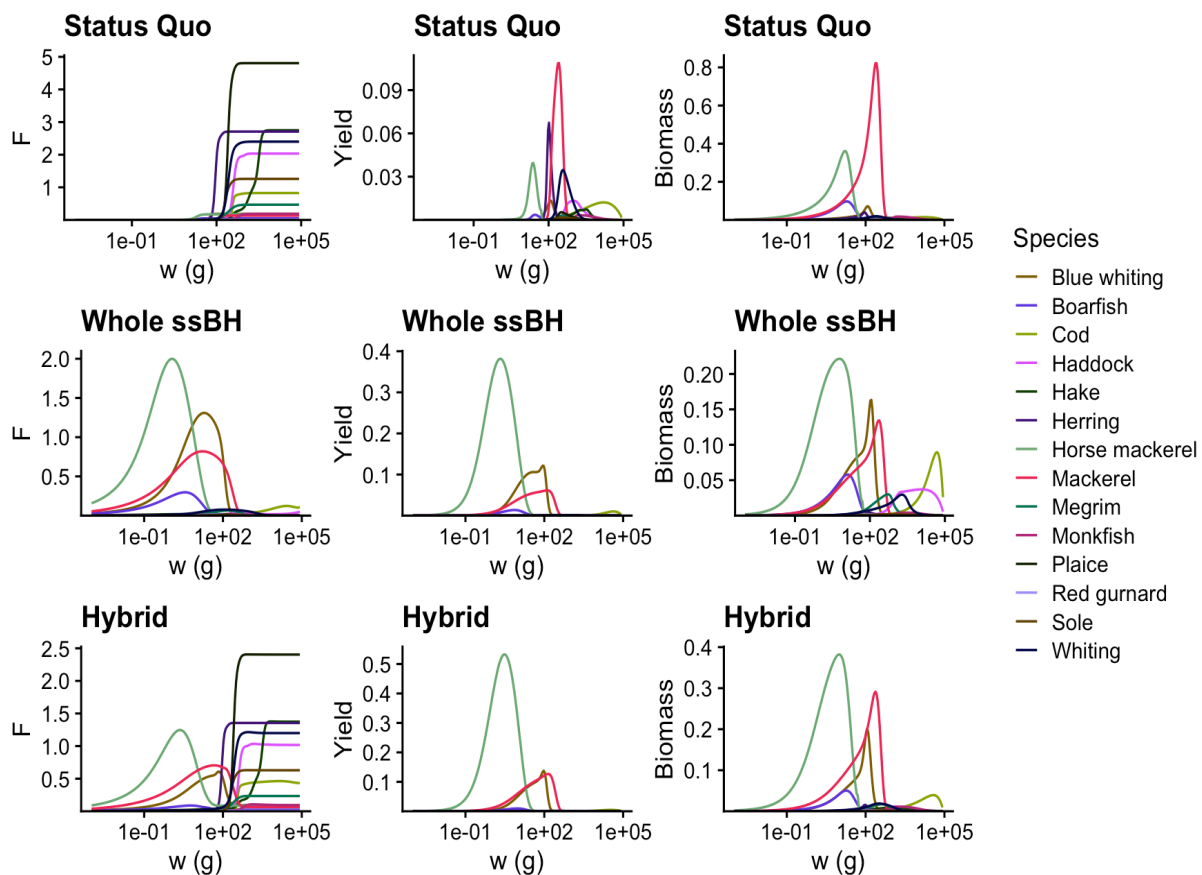


Figure 5: The size-species dependent fishing mortality (1/year), yield (grams/year/m²), and biomass (grams/year) at the steady-state of each of the key regimes.

fishing mortality here tapered off for larger individuals of a species, potentially protecting the largest individuals from fishing mortality. The yield profile was heavily dominated by much higher yields of small, highly productive pelagic species such as horse mackerel, mackerel, blue whiting, and boarfish, while yields for larger demersals were much lower than the status quo. Cod interestingly was harvested at much larger sizes in the ssBH regime than the status quo.

The steady-state of the hybrid model effectively bifurcated the fishing mortality, maintaining the exploitation of ssBH targeting small pelagics and status quo targeting large demersals. Despite this dual pressure, however, the total yield composition at the Hybrid steady-state remained dominated by small pelagics, causing it to resemble the whole-ssBH state more closely than the status quo in terms of total biomass harvested.

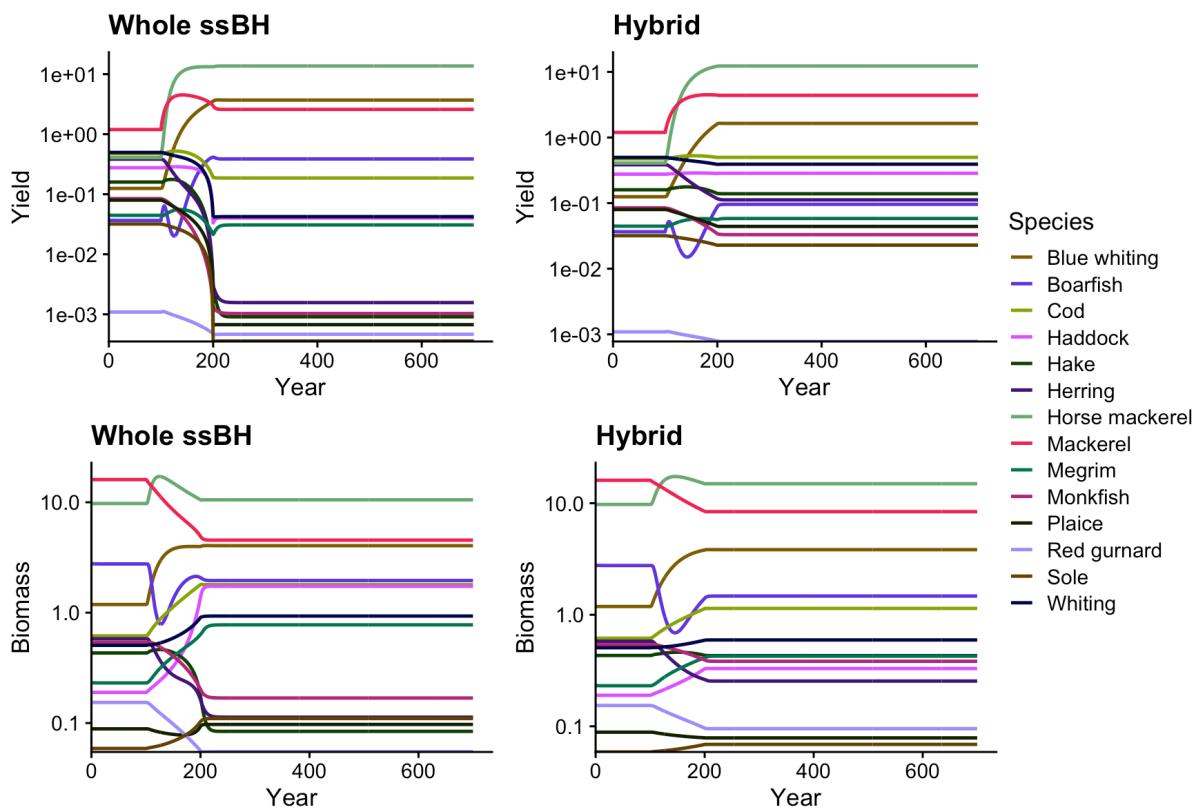


Figure 6: Biomass and yield of species from the status quo model projected forward in time and reallocated to the key hybrid fishing (right) and ssBH (left) fishing regimes. The status quo fishing regime is projected forward till $t=100$, fishing mortality is slowly allocated towards the new regime $100 < t < 200$, the regime is then run to steady-state in the new regime is run for 500 years till the steady-state value is evaluated at $t=700$.

4.4.2. SPECIES BIOMASS

The community biomass at these steady-states also exhibited significant restructuring. The hybrid and whole-ssBH steady-states had increased biomass for horse mackerel,

cod, blue whiting, sole, haddock, megrim, and whiting. The horse mackerel and blue whiting biomasses were particularly notable, as they maintained a higher standing biomass than the status quo despite supporting a manifold increase in yield. Conversely, the steady-state biomass of mackerel, boarfish, red gurnard, plaice, herring, and monkfish (Figure 7) in the hybrid and ssBH regimes were lower than the status quo steady-state.

Table 4: Species specific biomass and yield values for the status quo, best whole BH, and best overall regime (hybrid).

Metric	Species	Status Quo	Whole PssBH	Hybrid
Biomass	Horse mackerel	9.75	10.52	14.95
	Mackerel	16.05	4.54	8.40
	Blue whiting	1.19	4.04	3.84
	Boarfish	2.76	1.96	1.47
	Herring	0.58	0.11	0.25
	Sole	0.059	0.11	0.069
	Plaice	0.089	0.097	0.079
	Hake	0.43	0.084	0.43
	Red gurnard	0.15	0.055	0.095
	Cod	0.62	1.79	1.14
	Haddock	0.19	1.73	0.33
	Whiting	0.51	0.93	0.60
	Megrim	0.23	0.78	0.43
	Monkfish	0.54	0.17	0.38
Yield	Horse Mackerel	0.408	13.664	12.282
	Mackerel	1.195	2.589	4.391
	Blue Whiting	0.125	3.701	1.640
	Boarfish	0.037	0.387	0.096
	Herring	0.384	0.0016	0.112
	Sole	0.032	0.0004	0.023
	Plaice	0.080	0.0007	0.044
	Hake	0.160	0.0009	0.139
	Red gurnard	0.001	0.0005	0.001
	Cod	0.473	0.186	0.498
	Haddock	0.276	0.040	0.285
	Whiting	0.497	0.004	0.392
	Megrim	0.045	0.031	0.058
	Monkfish	0.085	0.001	0.033

4.5. SENSITIVITY ANALYSIS

4.5.1. MORRIS SENSITIVITY ANALYSIS

A Morris analysis identified the encounter coefficient (E_i) as the most influential parameter for status quo steady-state outcomes, particularly for mackerel, horse mackerel, boarfish, herring, whiting, cod and hake (Appendix 7.4).

4.5.2. RSA

RSA varied E_i by 10% for the aforementioned species across status quo, hybrid, and whole-ssBH regimes. The hybrid regime outperformed the status quo in M CCS in 94% of runs. Low intensity whole-BH regimes ($c=0.1 - 0.4$) also generally exceeded the status quo (79-91% of runs), whereas high-intensity whole-ssBH ($c=10$) underperformed it in 100% of runs.

In terms of highest M CCS within runs, the best-performing regime was most often whole BH at $c = 0.2$ (34%), followed by the hybrid regime (25%), $c=0.3$ (23%), with status quo, $c = 0.4$ or 0.1 rarely ranking best and $c = 1$ or 10 never ranked highest. Although the tested hybrid configuration did not consistently dominate all alternatives, other hybrid parameterisations may perform better.

Decomposing the M CCS showed broadly consistent patterns across individual metrics proving overall model conclusions were robust. Yield exceeded the status quo in 100% of hybrid runs, 98-100% of low intensity ssBH runs ($c<0.4$), and 91% of high-intensity whole-ssBH runs. B and SSB were higher than status quo at $c = 0.1$ in 84% and 86% of runs, respectively, but were generally lower or inconsistent at higher intensities. The hybrid regime reduced B in 60% of runs and SSB in 100%.

N was higher than status quo in 95% of runs at $c = 0.1$, but this declined steadily with increasing fishing intensity, falling to 3% at $c = 10$. The PBLFY was consistently lower under all alternative regimes relative to the status quo. However, BLFY was higher under hybrid regimes, whereas all whole-ssBH regimes produced lower BLFY in 100% of runs. BLFS was higher for all regimes except $c = 10$, while PBLFS was consistently higher across all ssBH regimes relative to the status quo.

Comparisons between low- and high-intensity whole-ssBH regimes confirmed the robustness of declines in Y, N, B, SSB, and BLFS at higher fishing intensities v of whole-ssBH regimes. Across all runs, B, SSB, and BLFS were higher at intensities between 0.1 and 0.4 compared to intensities of 1 or 10. Y and N, maintained the previously observed nonmonotonic relationship. Whole-ssBH at $c = 0.4$ produced higher yield than $c = 1$ and 10 in 99% of runs, while $c = 0.1, 0.2, 0.3$ produced higher yields in 3%, 59% and 91% of

runs, respectively. Similarly, N was higher at $c = 0.3$ and 0.4 in all runs relative to $c=1$ and 10 , while $c = 0.1$ and 0.2 exceeded those higher intensities in 90% and 98% of runs, respectively.

5. DISCUSSION

5.1. SSBH EFFECTS

The model revealed that the status quo regime in the Celtic Sea concentrated fishing mortality on large demersals, whereas ssBH regimes redistributed this exploitation towards small pelagics. This shift in exploitation structure led to a substantial increase in total yield derived from small pelagic biomass, resulting in consistently higher Y and lower PBLFY across all ssBH regimes relative to the status quo. In addition, it promoted the persistence of large fish within the system, reflected in higher PBLFS across all ssBH regimes and higher BLFS values in low intensity ssBH regimes.

The observed redistribution of fishing mortality was consistent with previous balanced harvesting studies and was theoretically expected, given the high biomass and productivity of low-trophic-level fish within marine ecosystems (Peters, 1986; Kolding et al., 2016a). These findings align with balanced harvesting and size-spectrum studies showing that broader exploitation across size classes can increase aggregate yields while reducing pressure on large-bodied species (Law et al., 2012; Jacobsen et al., 2014; Kolding et al., 2016c; Garcia et al., 2012; Law, 2007; Hixon et al., 2014).

Other metrics including B, SSB, N, PBLFS, and BLFY, varied substantially with both the weighting and fishing intensity of ssBH regimes. Whole-ssBH regimes at low fishing intensities produced the highest B, SSB, and N values, supporting the potential ecological benefits of balanced harvest (Zhou et al., 2019). However, although these low-exploitation regimes generated higher yields and improved ecosystem health relative to the status quo, they also produced the lowest BLFY values across all regimes, reflecting reduced fishing pressure on larger fish and, as such, potentially lower economic viability (Asche et al., 2014).

In contrast, hybrid regimes with low fishing intensity and low weighting were able to maintain high B, SSB, and N values while simultaneously increasing BLFY relative to the status quo. This indicates that hybrid regimes may better balance ecological and fishery objectives than whole-ssBH approaches. These outcomes arise because hybrid configurations retain reduced but ongoing status quo exploitation of demersal species while simultaneously shifting additional fishing pressure towards small pelagic species.

5.2. OPTIMAL REGIMES

The optimal regimes in this study were identified by equally weighting conservation and economic metrics to evaluate performance across all objectives. Under this weighting, the optimal regime was a hybrid configuration that increased yields by approximately five-fold, increased the contribution of large fish to catches, increased the abundance of large fish within the system, and maintained overall biomass comparable to the status quo. However, this regime also resulted in a substantial reduction in total SSB, highlighting a key trade-off between reproductive capacity and fishery objectives.

In practice, management priorities are unlikely to weigh all objectives equally, and certain metrics may be prioritised over others depending on the fishery context. For example, balanced harvest activists aim to minimize adverse impacts on the size and species composition of aquatic ecosystems, while maintaining or increasing sustainable yield, without prioritising yield maximisation at the expense of ecosystem health (Bundy et al., 2005; Zhou et al., 2010, 2015; Law et al., 2012, 2013, 2016; Garcia et al., 2012, 2016a; Jacobsen et al., 2014, 2017; Charles et al., 2016; Kolding et al., 2016b; Zhou et al., 2019). Within this context, the optimal regime identified in this study may not represent a true management optimum, as alternative weightings may better reflect stakeholder objectives.

In this study, regimes with lower ssBH intensity (~ 0.1 - 0.2) and intermediate weighting (~ 0.4 - 0.5) performed well across multiple objectives, maintaining or improving N, B, SSB, BLFS, BLFY, and Y relative to the status quo. These results suggest that hybrid configurations can simultaneously support ecological structure and fishery productivity, when maximum yield is not the sole management priority.

In systems where food provisioning is the primary objective, such as small-scale inland fisheries (i.e. lake Kariba, Tonlé Sap, and fisheries in the East China Sea), higher exploitation and higher weightings of ssBH regimes may be of interest (McCann et al., 2016; Szuwalski et al. 2017; Kolding et al. 2016c). In such systems, high yields have historically been associated with high fishing pressure and reduced predator biomass, enabling greater exploitation of small-bodied species (Jennings & Collingridge, 2015 ; Zhou et al., 2019). In this context, balanced harvesting regimes optimized for yield may offer a potential pathway to maintaining high food production while reducing the degree of structural simplification typically associated with predator depletion.

Overall, while the precise optimal regime will depend on how management objectives are weighted, the results indicate balanced harvesting frameworks can potentially support a range of conservational and fishery goals when appropriately configured.

5.3. FISHING INTENSITY

Despite the density-dependent formulation of fishing mortality, high fishing intensities under balanced harvesting shifted the system away from natural structure by homogenizing species biomasses and causing substantially lower B, SSB, N and BLFS. Ecologically desirable fishing intensities were, therefore, revealed to be those below 0.5.

5.4. PROPORTIONS

Under ssBH regimes, proportional metrics such as PBLFY and PBLFS were misleading, as increases in these metrics did not necessarily indicate recovery of large fish, as commonly assumed in stock assessments (Shephard et al., 2013; Shephard et al., 2012), but were instead driven by declines in small pelagic biomass. This highlights the need to interpret balanced harvesting outcomes using absolute measures rather than proportional indicators.

5.5. ASYMMETRIC REGIME COMPARISONS

It is important to note that status quo Celtic Sea fishing regimes do not accurately reflect current fisheries management recommendations (Salomon and Holm-Müller, 2013; ICES, 2025; Froese et al., 2008). Therefore, comparisons of ssBH with the status quo do not represent a direct contrast between alternative management strategies. Instead, this framework measures an optimized, ideal regime against a historically flawed baseline, creating an asymmetric comparison. Other studies have compared status quo management ideals to balanced harvest; however, they have used the balanced harvest productivity rule, reducing model comparability (Perez, 2019).

5.6. MODEL LIMITATIONS

All ecological models are simplifications of highly complex systems, and despite the fact that the size-spectrum approaches here avoid some of the major limitations associated with traditional fisheries models, they are no exception (Andersen et al., 2016; Pennino et al., 2025).

5.6.1. FISH ASSEMBLAGE

While modelling did represent background mortality and encounter to include non-explicitly modelled species, the fish assemblages were composed of only a subset of

the species present, and did not explicitly consider species of special concern for conservation, such as seabirds, marine mammals, and benthic fauna, all of which can be heavily impacted by fishery decisions (Cury et al., 2011; Pikitch et al., 2012).

5.6.2. *RESILIENCE AND STOCHASTICITY*

Resilience to fishing was a fundamental parameter in assessing ecosystem effects of high fishing pressure but was not tuned to empirical information; instead, the reproduction level, which represents such resilience, was set to a value of 0.8 for all species. While this parameter fixing allowed for model stability it may have reduced environmental realism and species predictability

Additionally, because dynamic resource interactions were omitted in favor of fixed background encounter and mortality rates, the model operated under a strict equilibrium assumption. Thus, any increases in species which were heavily reliant on the external encounter would have food automatically provisioned to them, bypassing the ecological reality where an expanding population would deplete its own forage base and trigger density-dependent starvation. Furthermore, the steady-state modelling approach and static nature of the external encounter meant that time-varying environmental productivity was not modelled. Consequently, the characteristic ‘boom-and-bust’ cycles of pelagic populations were not recreated. This decoupled their population dynamics from climate-driven productivity shifts, artificially forcing recruitment to be dictated entirely by a rigid stock-recruitment relationship, which contrasts with empirical evidence showing that forage fish fluctuations are primarily driven by environmentally induced recruitment pulses (Szuwalski et al., 2019).

Furthermore, assuming the Celtic Sea status quo represents a steady-state is potentially ecologically problematic. Official stock assessments indicate severe historical declines in key commercial species, with ICES repeatedly issuing zero-catch advice for cod and whiting due to populations falling below critical biomass thresholds (ICES, 2025a). Ecosystem models have further demonstrated that the Celtic Sea may currently be in an overfished state (Hervann and Gascuel, 2019; Perez, 2019). Consequently, treating the status quo as an equilibrium state, to allow simplified parameter tuning, failed to capture an ecosystem that is currently in a state of transition and overfished depletion. In addition to steady-state assumptions, models assumed ecosystems were non-seasonal and well-mixed.

For these reasons some conclusions may be model dependent, especially those concerning the effects of increased pelagic exploitation.

5.7. PELAGIC EXPLOITATION CONCERNS

Increased fishing of pelagics, as suggested by ssBH regimes in this study, remains highly controversial in fisheries science. Pelagic forage species occupy low trophic levels and play a critical role in marine ecosystems by transferring production from plankton to larger predatory fish, marine mammals, and seabirds. Previous studies have shown that depletion of forage fish can trigger trophic cascades affecting upper trophic levels (Jahnke et al., 2004; Frederiksen et al., 2005; Crawford et al., 2007; Smith et al., 2011). Similarly, Jennings and Collingridge (2015) suggested that increasing exploitation of lower trophic levels could increase total harvest potential but came at the expense of predator persistence due to reduced prey availability. Together, these studies demonstrate that heavy exploitation of low trophic level species, such as that produced under the ssBH regimes, may be extremely risky and should be approached cautiously.

Despite these concerns, the results of this study did not indicate widespread predator collapse or extinction under the ssBH regimes, even though fishing pressure was redistributed strongly towards pelagics. In contrast, overall pelagic biomasses increased relative to the status quo. Substantial increases were observed in species such as horse mackerel and blue whiting, likely due to reductions in mackerel, which preyed on them. Increased pelagic biomass may also have resulted from harvesting pelagics in closer proportion to their production and exploiting them at smaller sizes, potentially increasing overall productivity and turnover.

However, these outcomes should be interpreted cautiously. The increase in pelagic biomass despite elevated fishing mortality may be an artefact of model assumptions as discussed above, specifically regarding artificial resilience and recruitment dynamics (Essington et al., 2015). Pelagic species are highly sensitive to environmental variability, and their recruitment can fluctuate substantially between years (Pikitch et al., 2012). Historical collapses of highly exploited pelagic stocks, such as the Peruvian anchovy fishery, demonstrate how rapidly forage fish populations can decline when periods of poor recruitment coincide with high fishing pressure (Idyll, 1973). Such collapses can produce severe ecosystem-wide consequences, particularly when predator populations remain high (Pinsky et al., 2011).

Therefore, these results should be verified against other methods that have used the production rule of balanced harvest, which do not make steady-state assumptions and allow for environmental stochasticity. For example, Atlantis models may more accurately represent natural boom and bust cycles of forage fish. While an example exists of an Atlantis model in the Norwegian and Barents Sea (Nilsen et al., 2020), it

utilized a productivity balanced harvest rule, rather than a production rule and is, therefore, not directly comparable.

5.8. FUTURE WORK

Overall, this study highlights the potential ecosystem-wide impacts of shifting pressure towards ssBH from current fishing methods. While these results present a compelling case for these alternative harvest strategies, further investigation is required to test their robustness and viability.

While this study included a sensitivity analysis which supported modelled results, sensitivity analysis was focused on parameters tuned during the data-fitting process. Future work could expand this scope to explore untuned parameters. Specifically comprehensive sensitivity trials should be run across all hybrid regimes to determine if a specific hybrid configuration consistently outperforms other ssBH strategies under varying parameter spaces. Future modelling should also systematically vary the fixed reproduction level parameters to explicitly test how species-specific resilience to fishing pressure modifies the core conclusions.

Finally, because this study focused on abstract harvest theory, it deliberately bypassed the practical complexities of real-world implementation. Future work must bridge this gap by incorporating technological and operational constraints (Zhou et al., 2019).

Specifically, future frameworks should simulate realistic gear strategies and mesh size selectivity required to target precise size spectra. Additionally, introducing time-lags and inaccuracy in production and biomass assessments into the ssBH harvest applied would help evaluate whether the method would be effective under inevitable management inefficiencies (Ward et al., 2023). Crucially, the operational realities of mixed-fisheries, such as mixed removals and management constraints of choke species, must be explicitly integrated (Mortensen et al., 2018). Ultimately, operationalizing balanced harvest requires moving from idealized biomass-extraction models to logistically implementable frameworks and observing if such a framework could result in the same desired outcomes of the harvest theory.

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7. APPENDIX

7.1. DATA PROCESSING

7.1.1. *MATCHED DATA*

Data sources and ecological estimates included Lauria (2016), Scientific, Technical and Economic Committee for Fisheries (STECF) Fisheries dependent information (FDI) landings (European Commission, 2024), DATRAS survey data (ICES, 2025b), and data from Silva et al. (2024).

All data unless otherwise defined is taken from the Celtic Sea Area defined as ICES areas 27.7.b, 27.7.c, 27.7.e, 27.7.f, 27.7.g, 27.7.h, 27.7.j, and 27.7.k, covering a total area of 617499.7km², and was collected between 2012 to 2024. When data within the defined area and time period was insufficient, data outside of this year set and area were included.

7.1.2. *SURVEY DATA*

Fisheries-independent survey data were sourced from the DATRAS (Database of Trawl Surveys) online repository using the R package icesDATRAS (Millar et al., 2023). Data were extracted for the period 2012-2024 within the defined Celtic Sea ICES areas. The data included nine key surveys: SCOROC (Scottish Rockall Groundfish Survey), BTS (Offshore Beam Trawl Surveys), DYFS (Inshore Beam Trawl Surveys), IE_IGFS (Irish Ground Fish Survey), FR_CGFS (French Channel Groundfish Survey), IE_IAMS (Irish Anglerfish and Megrim Survey), SP_PORC (Spanish Porcupine Bottom Trawl Survey), EVHOE (French Southern Atlantic Bottom Trawl Survey), and NS_IBTS (North Sea International Bottom Trawl Survey). For each survey all available data types were retrieved, including haul meta data (HH), species age-based information (CA), and species length-based information (HL).

7.1.2.1. ICES Area and Rectangle Mapping

Spatial information in these surveys was provided via ICES rectangles. To subset data to the desired Celtic Sea Areas, these rectangles were mapped to broader ICES areas using shapefiles obtained from Marine Regions (Marine Regions, n.d.).

Spatial processing was conducted using the R package sf (Pebesma et al., 2026). Geometries were validated and reprojected to the ETRS89 Lambert Azimuthal Equal

Area coordinate reference system (EPSG:3035), following the Infrastructure for Spatial Information in Europe (INSPIRE) recommendations for European statistical mapping (Spatial Reference, 2026).

A spatial intersection was performed to identify the overlap between ICES rectangles and study areas. To avoid including data from rectangles that only shared boundaries with the study area, a 1% minimum overlap threshold was applied. For each rectangle in shape files a string of its respective overlapping ICES areas was assigned. To maintain strict spatial consistency, data were only retained for ICES rectangles that were contained entirely within the predefined Celtic Sea areas. If a rectangle overlapped with any area outside of the study boundaries, its data were excluded to prevent inclusion of data from external areas into the model.

7.1.2.2. Length Distribution

The HL dataset did not provide area codes and thus was joined with a subset of the CA dataset containing the necessary area assignments. These were harmonized by matching the survey identifiers: survey type, quarter, country, ship, gear, sweep length, station number, haul number, year, and species code. The resulting dataset was filtered to include the modelled fourteen species. Length classes were all standardized to cm and were rounded down to the 1 cm size length bins (ie: 24.5 into 24 cm). Finally, data was aggregated to length and species, with catches in number of individuals summed.

7.1.2.3. Stratified Age at Length data

The CA was joined to the HH metadata to append the specific month and day of each survey. This was necessary because the CA datasets lacked the required temporal information for matching the size at age data. Records were harmonized using the same survey identifiers, and latter filtering and length class standardization described above.

Maturity values were binarized from a multi-stage scale into a simplified classification of immature and mature individuals. The data were then aggregated by survey, quarter, species, year, month, length, age, and maturity with catches in number of individuals summed. For any species that had less than 1000 observations, ranges were expanded. Specifically, Red Gurnard data were expanded to include areas 27.7.a, 27.7.d and Hake data were expanded to include 27.4.b, 27.4.c, 27.7.a, and 27.7.d. Boarfish ages were not present in the DATRAS dataset and were sourced from Silva et al. (2024).

7.1.2.4. Boarfish Data

While the age-at-length data sourced from DATRAS follows a length-stratified sampling protocol, the supplemental data for Boarfish provided by Silva et al. (2024) was originally collected via random sampling. To ensure methodological consistency across the multi-species framework and to maintain compatibility with the established tuning and likelihood-matching protocols, the Boarfish dataset was post-hoc length-stratified.

This standardization was achieved by first assigning temporal quarters (Q1-Q4) to each record to match DATRAS conventions. Subsequently, for each unique combination of sampling date and 1 cm length class, a maximum of 10 individuals were randomly subsampled without replacement. These stratified records were then re-aggregated to represent the frequency of individuals per age within each length class. This data was then added to the processed age at length data detailed above and used in model tuning.

7.1.2.5. Calculating w_{mat} and w_{mat25}

To characterise the shape of species-specific maturity curves within the model, empirical estimates of weight at maturity parameters (w_{mat} , and w_{mat25}) were derived where data were available. These empirical estimates were used in place of literature-derived maturity (l_{mat}) values and allowed greater consistency with the weight-based formulation of mizer.

Maturity data were available for Red Gurnard, Cod, Megrim, Haddock, Hake, Whiting, Plaice, Mackerel, Sole and Boarfish. For all species except Boarfish, maturity observations were length-stratified. In these cases, data were grouped into discrete length classes, and within each class the number of maturity individuals and the total number of individuals were calculated to generate binomial counts. For Boarfish, individual-level (unstratified) observations were available and retained without aggregation.

Table A1: Maturity Conversion table

Original Values	New Value
2, 3, 4, 5, 52, 53, 54, 55, 62, 63, 64, 65, B, Ba, Bb, C, Ca, Cb, D, Da, Db, E, II, III, IV, IX, M, R1_2, R1_3, R1_4, R2_4, R2_6, R2_8, RF2, RF3, RF4, RF5, RF6, S2	Mature
S3, S4, S5, S6, V, VI, VII, VIII, X	
1, 51, 61, A, I, R1_1, R2_2, RF1, S1	Immature
-9	NA
6, 66, F	NA

Where individual weight-based measurements were available, these were used directly. Where only length data were available, individual weights were estimated using species-specific allometric length-weight relationships.

A binomial generalized linear model (GLM) with a logit link function was fitted to estimate maturity as a function of long-transformed weight:

$$\text{logit}(p) = a + B * \log(w) \quad \text{Eq. A1}$$

where p was the probability of an individual being mature.

From the fitted model, the weight at 50% maturity (w_{mat}) and 25% maturity (w_{mat25}) were derived analytically as:

$$w_{mat} = \exp\left(-\frac{a}{B}\right) \quad \text{Eq. A2}$$

$$w_{mat25} = \exp\left(\frac{\log\left(\frac{1}{3}\right) - a}{B}\right) \quad \text{Eq. A3}$$

This formulation was consistent with the maturity functions used in mizer at allowed direct integration of these empirical estimates into the model.

7.1.3. LANDINGS

In the model, fishing yields ($Yield^*$, $grams/m^2/year$) were parameterized based on the total biomass of dead fish resulting from fishing activities. This approach accounts for the cumulative mortality from both landings and discards, rather than solely the landed biomass. By defining yield as total removals, the model achieves a more accurate representation of the true mortality pressure on the ecosystem and the subsequent state of the fish community.

To represent this pressure, two distinct data frames were generated: one containing the total biomass mortality per species and gear, and a second containing the normalized size-frequency distributions for these removals. This reconstruction required the integration of three distinct data streams, with annual landed biomasses being sourced from Lauria (2016)/ICES stock assessments, and size distributions and biomass discard rates were calculated from the STECF FDI database (2013-2023).

To ensure spatial consistency, all supplemental STECF FDI data were subjected to a strict spatial filter. In which only fishing trips occurring entirely within the predefined

ICES sub-areas were included. Records containing mixed spatial strings were excluded to maintain the biological integrity of the Celtic Sea signal.

7.1.3.1. Discard Rates and Total Yield* Calculation

Reconstructing the total biomass of dead fish required adding the biomass of fish killed through discarding to the total annual landed biomass. The species (i) and gear (g) specific total yield ($Yield^*_{i,g}$) was calculated using the following equation:

$$Yield^*_{i,g} = Yield_{landed,i,g} + Yield_{landed,i,g} * R_{D,i,g} * 0.5 \quad \text{Eq. A4}$$

where $R_{D,i,g}$ was the species and gear-specific discard rate. Despite the high variability in discard survival rates for species, gear types and sizes, a 50% post-release mortality was assumed for all discarded biomass (ICES, 2020).

The discard rates $R_{D,i,g}$ were derived from the STECF FDI catch records. Commercial confidential records were removed from the dataset, and the remaining data were aggregated by gear category, country and target assemblage. The discard rate was defined as the ration of total discard biomass ($B_{discard}$) to total landed biomass (B_{landed}):

$$R_{D,i,g} = \frac{\sum B_{discard,i,g}}{\sum B_{landed,i,g}} \quad \text{Eq. A5}$$

7.1.3.2. Size-Specific Mortality Distributions

The STECF FDI data provided length-frequency information for both landings and discards. Two normalized resulting species and gear specific size distributions were created one for the landing (Φ_L) and the other for the discards (Φ_D). To align these with the calculated biomass mortality, a weighted normalized distribution was created for each species-gear combination (ICES, 2020).

The total species and gear-specific dead normalized distribution ($\Phi_{Total,i,g}$) was calculated as:

$$\Phi_{Total,i,g} = \Phi_{L,i,g} \left(\frac{1}{1 + R_{D,i,g} * 0.5} \right) + \Phi_{D,i,g} \left(\frac{R_{D,i,g} * 0.5}{1 + (R_{D,i,g} * 0.5)} \right) \quad \text{Eq. A6}$$

For data-limited species Herring and Monkfish, size-frequency proxies were sourced from biologically adjacent ICES divisions (27.7.d and 27.3.1.20, respectively) to ensure robust mortality profiles.

7.1.3.3. Gear Categorization

A primary technical challenge was the discrepancy between the gear classifications used in Lauria (2016) and the STECF FDI repository. A mapping protocol was established to translate the high-resolution STECF gear codes to the broader functional fleet categories (Trawl, Gillnet, Pelagic, and Misc) required for model consistency.

A redistribution protocol was implemented to manage data gaps where biomass mortality was recorded for a gear from Lauria, but no corresponding size distribution was available. In these instances, the biomass was re-allocated to the most functionally similar gear for that species. This ensured that the total system biomass removals remained accurate while utilizing the most representative size-selective signals available.

Table A2: Gear Conversion Table

Final Gears Categories	STECF Definition	Lauria Definition
Trawl	OTT, OTB, OTB-OTT-PTB, TBB	UK Otter trawl, UK Pair trawl, UK Beam trawl, Ireland Otter trawl, Ireland Pair trawl, Ireland Beam trawl, France Otter trawl, France Pair trawl, France Beam trawl, Spain Otter trawl, Spain Pair trawl, Spain Beam trawl
Gillnet	GND-GNS-GNC, GNS, GTR, GTR-GTN	UK Gillnets, Ireland Gillnets, France Gillnets, Spain Gillnets
Pelagic	LHM-LHP, OTM, OTM-PTM, LLS, PS, PS-LA, SSC	UK Pelagic fleet, Ireland Pelagic fleet
Misc	All other	UK Pots and traps, UK Dredge, Ireland Pots and traps, Ireland Dredge, France Pots and traps, France Dredge, Spain Pots and traps, Spain Dredge, Other countries

7.1.4. PREDATION KERNELS

The predation kernel describes a predator's size-based feeding preference, defining the range of prey sizes it is capable of consuming. The sizes of prey a predator can consume relative to its own body size are assumed to be size invariant (Tsai et al., 2016). It can, therefore, be expressed as a function of the predator-prey mass ratio ($\frac{w'}{w}$), defining the preference of a predator of size w for prey of size w' . In the model, the feeding kernel ($\varphi_j\left(\frac{w'}{w}\right)$) was defined as a power-law feeding kernel:

$$\varphi_j\left(\frac{w'}{w}\right) = \left(\frac{w'}{w}\right)^{kernel_{exp}} \left(1 + \left(\frac{\exp(kernel_{ll})}{\left(\frac{w'}{w}\right)}\right)^{kernel_{ul}}\right)^{-1} \left(1 + \frac{\frac{w'}{w}}{(\exp(kernel_{lr}))^{kernel_{ur}}}\right)^{-1} \quad \text{Eq. A7}$$

Where the $kernel_{exp}$ was the exponent of the power law, $kernel_{ll}$ and $kernel_{ul}$ were the and shape of the left (rising) sigmoid, respectively, $kernel_{lr}$ and $kernel_{ur}$ was the location and shape of the right (falling) sigmoid.

The predation kernel parameters were determined through visual matching to stomach contents data sourced from ICES and CEFAS (available on the [gustavdelius/mizerStomach](#) GitHub repository) using the mizerStomach application. Parameters were chosen which simultaneously matched the prey biomasses and number densities across predator-prey mass ratios (Reum et al., 2018).

It is recognized that stomach contents data does not accurately reflect predation size preferences, as observed diets depend on prey availability within the ecosystem (Tsai et al., 2016). To reduce this bias, the dataset used spanned the years 1836 to 2016 and included observations from areas both within and beyond the defined Celtic Sea region. This broad temporal and spatial coverage allowed more accurate representation of the range of prey sizes each predator can consume, capturing the fundamental feeding capacity rather than context-dependent diet composition. Matching predation kernels to these data, rather than standardized values non-species-specific values, was important for improving the ecological realism and predictive performance of the model (Murphy et al., 2023).

7.1.5. INTERACTION MATRIX

The interaction matrix contains the biological information describing species interactions and can be estimated via spatial overlap, predator's diet preferences, and prey vulnerability to predation.

In this study, the interaction matrix was derived from the diet matrix, which originated from the Ecopath model of Lauria (2016). The ecopath diet matrix contained proportions of each predator's diet derived from different ecological groups. These proportions were first converted into absolute consumption rates by multiplying the total consumption of a predator as estimated within the ecopath model.

For predator i , total consumption was given by:

$$Q_i = \int_{w_0}^{w_{max}} (E_i w^n) (1 - f_i(w)) N_i(w) dw \quad \text{Eq. A8}$$

and the corresponding diet contribution from prey j is:

$$D_{m,ij} = D_{mp,ij} Q_i \quad \text{Eq. A9}$$

where $D_{mp,i}$ represented the diet proportions from Ecopath diet matrix for predator i , and $D_{m,i}$ is the resulting absolute consumption of prey j by predator i .

To obtain the interaction matrix, the diet matrix was scaled by the encounter matrix. The encounter matrix was defined as:

$$E_{i,j} = \int_{w_0}^{w_{max}} (1 - f_j(w)) \gamma_j(w) \int N_j(w') \varphi_j\left(\frac{w'}{w}\right) w' dw' \quad \text{Eq. A10}$$

From this diet matrix and encounter matrices, the interaction strengths (θ_{ij}) required to reproduce the observed diets was calculated as:

$$\theta_{ij} = \frac{D_{m,ij}}{E_{i,j}} \quad \text{Eq. A11}$$

7.1.6. DIET MATRIX

The diet matrix originated from the Ecopath model developed by Lauria (2012,2016). While this Ecopath model included many of the species used in the present study, two focal species, red gurnard and boarfish, were not explicitly resolved and were instead, embedded within broader ecological groups (gurnards and small pelagics, respectively). The proportion of biomass attributable to these species within their groups was given by Lauria (2012) as 73% for red gurnard and 97% for boarfish. To preserve the original diet structure while separating these species, new species stanzas were introduced, and their contributions were partitioned from the parent groups by rescaling according to biomass.

The *reduceEcopathDiet()* function was then used to finalise the diet matrix. This function moved unmodelled groups into an “other” category and combined species represented by multiple stanzas (i.e. juvenile and adult stages) into a single-species stanza. When combining these stanzas, diet proportions could not simply be summed or averaged because they were defined relative to each group’s own total consumption, which can differ substantially between the split stanzas for a single species. Instead,

each group's diet was first scaled by its total consumption as estimated within the Ecopath model, converting proportions into absolute consumption rates. These split stanzas were then summed into a single stanza to represent the species and were subsequently re-normalised to proportions. This ensured that groups contributing more to the total consumption had a proportionally greater influence on the combined diet.

The final output was a predator-prey matrix giving, for each species, the proportion of its total consumption derived from each prey species and from an additional "other" category representing unmodelled components.

7.2. FUNCTIONS TABLE

Table A4: *Pre-existing mizer functions, descriptions and sources integral to model creation plan.*

Function	Description	GitHub repository
newAllometricParams()	Set up parameters for a model with allometric encounter and mortality rates.	<i>gustavdelius/mizerEcopath</i>
MatchBiomass()	Multiplies for each species the abundance density for all sizes by a factor to make total biomasses match observed biomasses.	<i>sizespectrum/mizer</i>
MatchYield()	The function adjusts species and gear specific catchability to match observed yields.	<i>sizespectrum/mizerExpiremental</i>
steadySingleSpecies()	Uses the growth and death rates to determine the steady-state abundances, setting single species to steady-state abundances.	<i>sizespectrum/mizer</i>
matchDiet()	Matches the diet matrix of the model to the observed diet matrix.	<i>gustavdelius/mizerEcopath</i>
makeInteracting()	Called by matchDiet, External encounter and mortality rates are decreased by the amount of predation encounter and mortality that is created by the interaction matrix.	<i>gustavdelius/mizerEcopath</i>
projectToSteady()	Run the full dynamics, but stop once the change has slowed down sufficiently, in the sense that the distance between states at successive time steps is less than a predetermined distance.	<i>sizespectrum/mizer</i>
getBiomass()	Calculate the total biomass of each species within a size range at each time step.	<i>sizespectrum/mizer</i>

getSSB()	Calculates the spawning stock biomass (SSB) through time of the species in the `MizerSim` class. SSB is calculated as the total mass of all mature individuals.	<i>sizespectrum/ mizer</i>
getYield()	This yield rate is given in grams per year. It is calculated at each time step saved in the MizerSim object.	<i>sizespectrum/ mizer</i>

7.3. MODEL PARAMETER VALUES

Table A5: Reproduction Parameters

Species	age_{mat}	w_{mat}	w_{mat25}	m	$erepro$	R_{max}
Blue whiting	2.04	84.41	69.591	0.75	6.591362e-01	8.254223e+01
Boarfish	3.78	10.57	4.522	0.70	3.875251e-01	8.814790e+01
Cod	4.68	675.39	301.958	1.50	2.962345e-03	4.650358e-02
Haddock	3.24	117.83	50.601	1.50	4.258901e-05	1.278639e-03
Hake	3.16	210.52	131.501	1.01	8.273190e-01	2.745387e+01
Herring	3.09	79.78	72.588	0.88	1.493775e-01	3.753024e+00
Horse mackerel	2.69	11.05	5.710	0.70	9.343926e-01	5.432064e+02
Mackerel	2.09	160.15	108.164	0.71	2.594689e-01	3.069331e+02
Megrim	2.45	69.48	48.412	1.13	1.693839e-02	2.274724e-01
Monkfish	4.50	1246.64	1175.467	0.92	8.193463e-03	2.744162e-01
Plaice	3.16	55.41	33.641	0.78	6.185221e-01	4.363338e+00
Red Gurnard	2.62	85.62	54.052	0.71	1.332904e-01	2.308535e+00
Sole	2.63	82.98	52.223	1.23	1.198125e-01	2.506988e-01
Whiting	1.70	68.59	44.792	1.26	2.404807e-02	5.606369e-01

Table A6: Predation Kernel Parameters

Species	$predation_{kernel}$	$kernel_{exp}$	$kernel_{ll}$	$kernel_{ul}$	$kernel_{lr}$	$kernel_{ur}$
Blue whiting	power_law	0.333	2.1	3.4	8.2	1.5
Boarfish	power_law	-0.017	3.6	1.9	10.6	2.4
Cod	power_law	-0.867	4.6	3.0	12.5	4.3
Haddock	power_law	-0.217	4.5	2.2	12.0	2.3
Hake	power_law	-0.317	4.4	3.8	4.5	10.0
Herring	power_law	-0.117	3.6	2.2	14.8	1.4
Horse mackerel	power_law	0.133	3.4	1.6	12.5	1.7
Mackerel	power_law	0.033	3.8	2.1	15.8	2.3
Megrim	power_law	-1.667	3.9	4.0	8.2	1.3
Monkfish	power_law	0.333	0.9	10.0	5.3	3.4
Plaice	power_law	-1.267	5.4	3.4	14.3	4.8
Red Gurnard	power_law	-0.767	4.2	2.4	10.1	2.8
Sole	power_law	-0.667	5.2	2.4	10.6	1.6
Whiting	power_law	-0.567	1.38	3.1	14.1	1.5

Table A7: Allometric Background Scaling Parameters

Species	n	p	d	q	h	k_s	E	μ_{mat}	d/g
Blue whiting	0.7	0.7	-0.3	0.75	33.625	4.708	13.45	1.95	0.0049
Boarfish	0.7	0.7	-0.3	0.75	11.425	1.600	4.57	1.08	0.0005
Cod	0.7	0.7	-0.3	0.75	64.150	8.981	25.66	0.87	0.3081
Haddock	0.7	0.7	-0.3	0.75	70.050	9.807	28.02	0.29	0.6345
Hake	0.7	0.7	-0.3	0.75	55.125	7.718	22.05	2.45	0.3452
Herring	0.7	0.7	-0.3	0.75	20.950	2.933	8.38	0.86	0.0452
Horse mackerel	0.7	0.7	-0.3	0.75	9.625	1.348	3.85	1.01	0.0058
Mackerel	0.7	0.7	-0.3	0.75	25.425	3.560	10.17	1.12	0.0008
Megrim	0.7	0.7	-0.3	0.75	18.275	2.559	7.31	0.69	0.1250
Monkfish	0.7	0.7	-0.3	0.75	39.575	5.541	15.83	0.64	0.3400
Plaice	0.7	0.7	-0.3	0.75	28.350	3.969	11.34	1.34	2.7010
Red Gurnard	0.7	0.7	-0.3	0.75	28.875	4.043	11.55	1.39	0.3290
Sole	0.7	0.7	-0.3	0.75	21.950	3.073	8.78	0.83	0.8413
Whiting	0.7	0.7	-0.3	0.75	35.475	4.967	14.19	1.19	0.1252

Table A8: Spawning and Annuli Parameters

Species	Spawning_mu	spawning_kappa	annuli_min_age	annuli_date
Blue whiting	0.29	2.43	1.75	0.92
Boarfish	0.58	1.60	0.00	0.22
Cod	0.33	7.05	1.27	0.06
Haddock	0.42	3.05	0.47	0.88
Hake	0.50	2.43	0.48	0.16
Herring	0.88	0.01	1.53	0.00
Horse mackerel	0.46	1.60	2.95	0.10
Mackerel	0.33	3.04	2.72	0.92
Megrim	0.21	4.67	0.87	0.11
Monkfish	0.17	1.60	0.84	0.16
Plaice	0.21	3.05	0.04	0.77
Red Gurnard	0.50	3.07	1.96	0.00
Sole	0.46	7.61	1.09	0.03
Whiting	0.37	1.38	0.87	0.17

Table A9: Gear Parameters

Species	gear	L_{50}	L_{25}	Q	$Y_{observed}$
Blue whiting	Misc	23.9	23.0	1.096478e-05	5.816488e-06
Blue whiting	Pelagic	21.3	20.6	1.819701e-01	1.185230e-01
Blue whiting	Trawl	23.4	22.0	5.370318e-03	2.910923e-03
Boarfish	Trawl	12.0	11.3	4.786301e-02	3.712090e-02
Cod	Gillnet	39.9	35.7	9.772372e-02	5.377759e-02
Cod	Pelagic	36.5	35.8	6.606934e-02	3.782194e-02
Cod	Trawl	34.2	33.6	6.606934e-01	3.958205e-01

<i>Haddock</i>	<i>Gillnet</i>	48.4	46.6	8.350414e-02	6.281697e-03
<i>Haddock</i>	<i>Trawl</i>	35.4	33.7	1.949845e+00	2.714282e-01
<i>Hake</i>	<i>Gillnet</i>	73.6	70.3	1.737801e+00	1.182773e-01
<i>Hake</i>	<i>Pelagic</i>	56.1	52.2	7.413102e-01	1.580560e-01
<i>Hake</i>	<i>Trawl</i>	36.8	34.9	2.691535e-01	6.710940e-02
<i>Herring</i>	<i>Gillnet</i>	23.4	23.3	7.673823e-03	9.599614e-04
<i>Herring</i>	<i>Pelagic</i>	22.9	22.1	2.519596e+00	4.009127e-01
<i>Herring</i>	<i>Trawl</i>	27.1	26.0	1.819701e-01	5.317706e-03
<i>Horse mackerel</i>	<i>Pelagic</i>	22.6	20.8	1.698244e-01	3.706210e-01
<i>Horse mackerel</i>	<i>Trawl</i>	9.3	8.9	2.398833e-03	1.985521e-02
<i>Mackerel</i>	<i>Gillnet</i>	27.7	25.8	1.819701e-03	1.141559e-02
<i>Mackerel</i>	<i>Misc</i>	30.3	29.6	1.096478e-03	5.099996e-03
<i>Mackerel</i>	<i>Pelagic</i>	24.6	23.8	1.174898e-01	9.534507e-01
<i>Mackerel</i>	<i>Trawl</i>	24.0	23.4	1.230269e-02	1.029672e-01
<i>Megrim</i>	<i>Trawl</i>	33.1	29.1	4.677351e-01	2.638248e-01
<i>Monkfish</i>	<i>Trawl</i>	35.9	33.1	1.862087e-01	8.404867e-02
<i>Plaice</i>	<i>Gillnet</i>	31.3	29.2	4.265795e-01	7.826767e-03
<i>Plaice</i>	<i>Misc</i>	33.1	31.6	9.772372e-02	1.537828e-03
<i>Plaice</i>	<i>Pelagic</i>	34.9	33.4	2.089296e-01	2.998630e-03
<i>Plaice</i>	<i>Trawl</i>	29.3	28.1	4.073803e+00	8.373198e-02
<i>Red gurnard</i>	<i>Gillnet</i>	31.8	29.6	2.290868e-04	6.018015e-06
<i>Red gurnard</i>	<i>Pelagic</i>	24.2	21.9	1.737801e-03	1.132225e-04
<i>Red gurnard</i>	<i>Trawl</i>	22.9	20.2	1.202264e-02	9.105086e-04
<i>Sole</i>	<i>Gillnet</i>	31.7	29.5	3.235937e-01	6.660909e-03
<i>Sole</i>	<i>Misc</i>	28.9	27.0	2.238721e-02	5.095585e-04
<i>Sole</i>	<i>Trawl</i>	27.8	26.5	9.120108e-01	2.139644e-02
<i>Whiting</i>	<i>Gillnet</i>	41.1	37.4	1.513561e-01	1.448022e-02
<i>Whiting</i>	<i>Pelagic</i>	31.8	29.0	5.495409e-01	1.337574e-01
<i>Whiting</i>	<i>Trawl</i>	32.5	30.6	1.698244e+00	4.388444e-01
<i>Boarfish</i>	<i>Survey</i>	5.5	5.3	4.406440e-07	1.000000e-06
<i>Red gurnard</i>	<i>Survey</i>	20.6	18.2	5.623413e-06	1.000000e-06
<i>Herring</i>	<i>Survey</i>	11.1	10.5	1.949845e-06	1.000000e-06
<i>Cod</i>	<i>Survey</i>	40.2	32.5	1.803951e-06	1.000000e-06
<i>Megrim</i>	<i>Survey</i>	20.1	18.9	5.128614e-06	1.000000e-06
<i>Monkfish</i>	<i>Survey</i>	15.3	12.9	1.905461e-06	1.000000e-06
<i>Haddock</i>	<i>Survey</i>	14.2	13.4	5.321487e-06	1.000000e-06
<i>Whiting</i>	<i>Survey</i>	11.1	10.1	2.187762e-06	1.000000e-06
<i>Hake</i>	<i>Survey</i>	10.6	9.5	3.630781e-06	1.000000e-06
<i>Blue whiting</i>	<i>Survey</i>	14.3	13.9	1.096478e-06	1.000000e-06
<i>Plaice</i>	<i>Survey</i>	15.6	5.1	1.862087e-05	1.000000e-06
<i>Mackerel</i>	<i>Survey</i>	6.8	6.5	6.456542e-08	1.000000e-06
<i>Sole</i>	<i>Survey</i>	16.6	14.7	1.621810e-05	1.000000e-06
<i>Horse mackerel</i>	<i>Survey</i>	12.6	11.7	1.445440e-07	1.000000e-06

Table A10: Misc Parameters

<i>Species</i>	w_{repro_max}	w_{max}	B_{cutoff}	$B_{observed}$	w_{min}	a	b	$alpha$
<i>Blue whiting</i>	1187.1	1154.3	97.1	0.444	0.001	0.0042	3.12	0.7
<i>Boarfish</i>	102.0	278.9	1.0	2.474	0.001	0.0191	2.82	0.7
<i>Cod</i>	82626.4	82626.5	693.7	0.550	0.001	0.0068	3.08	0.7
<i>Haddock</i>	1601.8	13545.4	110.0	0.179	0.001	0.0066	3.08	0.7
<i>Hake</i>	10044.4	23236.2	253.0	0.260	0.001	0.0047	3.12	0.7
<i>Herring</i>	684.0	733.4	1.0	0.564	0.001	0.0055	3.10	0.7
<i>Horse mackerel</i>	47.1	524.7	1.0	8.288	0.001	0.0018	2.96	0.7
<i>Mackerel</i>	577.4	1858.5	1.0	15.653	0.001	0.0065	3.07	0.7
<i>Megrim</i>	780.2	2861.8	56.9	0.194	0.001	0.0048	3.10	0.7
<i>Monkfish</i>	13291.6	84320.9	1969.7	0.290	0.001	0.0170	2.91	0.7
<i>Plaice</i>	14757.2	10711.8	52.9	0.050	0.001	0.0078	3.07	0.7
<i>Red Gurnard</i>	3516.9	3368.0	1.0	0.150	0.001	0.0079	3.05	0.7
<i>Sole</i>	1877.0	3343.4	1.0	0.058	0.001	0.0072	3.07	0.7
<i>Whiting</i>	3236.9	6197.8	70.0	0.412	0.001	0.0062	3.06	0.7

7.4. R CODE

All code for model construction and projection as well as the publicly available data files are available on the GitHub repository accessible through the following link: <https://github.com/jessicawestworth/mizerEcopathedits/tree/dissertation>

The README contains information of the files and models directly relevant to sections of the report and explains the installation of the package.

7.5. APPENDIX REFERENCES

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